

Almost-perfect non-local quantum computation requires high-dimensional resources

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Status of these notes. This is an almost-entirely AI-written exposition of extensions of [FGMS26] to approximate protocols. We have spent time understanding the main proof, but have not simplified, reconstructed, and rewritten these arguments as we have for the main paper. Several of the results also have AI-generated Lean proofs, with the assumptions and coverage described in Section 9; the sources are at <https://github.com/fspeelman/NLQC/>.

Abstract

The main paper proves that almost every bipartite unitary, already on two qubits, has no exact non-local quantum computation protocol with finite-dimensional resources. These companion notes give lower bounds on the resources needed as the approximation error ε tends to zero.

We charge the footprint $K = r m_A m_B$, the resource Schmidt number times the dimensions of the two simultaneous messages. For n qubits per party, universal approximation of unitaries or rank-one projective measurements requires $\log_2 K \geq 2n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$. For almost every fixed target, below a target-dependent error threshold, the bound is $n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$. The errors are Choi entanglement infidelity for unitaries and average joint-label failure on basis inputs for measurements; the bounds also hold for normalized diamond and worst-case total-variation error, respectively. We obtain weaker precision bounds when classical messages are free, including for measurement localization, and extend the fixed-dimension lower bound to almost every diagonal gate.

The proof controls the overlap of compressed encoder and decoder isometries. Its derivative is tangent to the local-unitary orbit up to a leakage term, and a dimension-uniform tube estimate bounds the volume of approximately reachable targets. Parts of the argument are formalized in Lean 4, conditionally on three stated results of real algebraic geometry.

1 Introduction

In non-local quantum computation (NLQC), two parties apply a joint operation to their quantum inputs using a shared entangled state and one simultaneous exchange of messages. The main paper [FGMS26] proves that Haar-almost every bipartite unitary, already on two qubits, has no exact protocol with finite-dimensional resources; the same holds for rank-one projective measurements whose outcome both parties must learn. Every unitary nevertheless has arbitrarily accurate finite-resource approximations [Vai03, BCF⁺14, BK11]. These notes ask how fast the required resources grow as the error tends to zero.

We charge a protocol its *footprint* $K = r m_A m_B$, where r is the Schmidt number of its resource state and m_A, m_B are the message dimensions. Classical and quantum messages are charged alike; all private registers are free. Our bounds constrain Schmidt numbers and message dimensions, not entanglement entropy. The errors are the Choi entanglement infidelity δ_{Choi} for unitaries and the average probability δ_Φ that the parties do not both output the correct label on a uniformly sampled basis input for measurements. Normalized diamond distance and worst-case total-variation distance bound these errors from above (Section 2.2).

1.1 Main results

Let $d = 2^n$ be the local input dimension, with $n \geq 1$. The proofs also hold for every integer $d \geq 2$. Constants $c, C > 0$ and $O(1)$ are universal, independent of d , the footprint and the error. Write $K_\varepsilon(T)$ for the least footprint that approximates a target T with error at most ε . For any fixed target without an exact finite-resource protocol, compactness already gives $K_\varepsilon(T) \rightarrow \infty$ (Proposition 2.2); our first theorem gives rates for almost every target.

Theorem A (Haar volume; Theorem 5.5 and Corollaries 5.6 and 5.7). *Let $n \geq 1$ and $0 < \varepsilon \leq 1/2$.*

- (i) *Volume. For $K \geq d^2/4$, the unitaries that some protocol of footprint at most K implements with $\delta_{\text{Choi}} \leq \varepsilon$ have Haar measure at most*

$$\exp(Cd^2 K^2) \varepsilon^{\kappa_u/2}, \quad \kappa_u = (d^2 - 1)(d^2 - 3).$$

The same holds for rank-one projective measurements with $\delta_\Phi \leq \varepsilon$, with the invariant measure on bases and exponent $\kappa_m = (d^2 - 1)(d^2 - 2)$.

- (ii) *Universal approximation. A footprint that suffices for every unitary, or for every rank-one projective measurement, satisfies*

$$K \geq c d \sqrt{\ln(1/\varepsilon)}, \quad \text{that is,} \quad \log_2 K \geq n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1).$$

- (iii) *Almost every target. For Haar-almost every $U \in \mathcal{U}(d^2)$ there is a threshold $\varepsilon_0(U) > 0$ such that $\log_2 K_\varepsilon(U) \geq n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$ for all $0 < \varepsilon < \varepsilon_0(U)$. The same holds for almost every rank-one projective measurement.*

The condition $K \geq d^2/4$ in part (i) holds whenever the footprint approximates **SWAP** or any basis measurement at error at most $1/2$ (Lemma B.1). The exponents are the codimension bounds from the exact argument. For universal approximation, we can improve the dimension factor by considering targets near **SWAP**, or near a maximally entangled basis.

Theorem B (Universal approximation; Corollaries 6.4 and 6.8). *Let $n \geq 1$ and $0 < \varepsilon \leq 1/2$.*

- (i) *A footprint that suffices for every unitary satisfies*

$$K \geq c d^2 \max\{1, \sqrt{\ln(1/\varepsilon)}\}.$$

- (ii) *A footprint that suffices for every rank-one projective measurement satisfies*

$$K \geq c \max\{d^3, d^2 \sqrt{\ln(1/\varepsilon)}\}.$$

In both cases, $\log_2 K \geq 2n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$.

The distinction is between universal and generic bounds. Theorem A controls the whole target group using $O(d^2 K^2)$ witness coordinates. The neighbourhoods used for Theorem B force both message dimensions to be of order d and permit smaller frozen environments, reducing this count to $O(K^2)$. Every universal budget must cover these neighbourhoods, but they do not give a stronger whole-group generic bound. The elementary floors are $K \geq d^2(1 - \varepsilon)$ for universal unitary approximation and $K \geq d^3(1 - \varepsilon)^2$ for universal PVM approximation; almost every fixed target has the corresponding integer floor d^2 or d^3 below a target-dependent threshold.

The divergence with precision survives when classical communication is free.

Theorem C (Free classical communication; Lemma 7.2 and Corollaries 7.3 and 7.4). *Suppose classical messages are free, possibly with continuous alphabets, and charge only the quantum footprint $K_q = r m_A^q m_B^q$, where m_A^q, m_B^q are the dimensions of the quantum parts of the messages. For $0 < \varepsilon \leq 1/2$, a quantum footprint that suffices for every unitary, or for every rank-one projective measurement, satisfies*

$$\log_2 K_q \geq \frac{1}{10} \log_2 \ln(1/\varepsilon) - \frac{2}{5}n - O(1),$$

and in addition $K_q \geq d^2(1 - \varepsilon)$ for unitaries and $K_q \geq d(1 - \varepsilon)$ for measurements. For almost every fixed unitary or measurement, below a target-dependent error threshold, the same precision bound holds with $\frac{3}{5}n$ in place of $\frac{2}{5}n$, and in addition $K_q \geq d^2$ for unitaries and $K_q \geq d$ for measurements. With only classical messages (local operations and simultaneous classical communication, LOSCC), $K_q = r$ is the Schmidt number of the resource; in particular, localizing the measurement in a Haar-random basis without communication, to failure probability ε below a threshold, requires a resource of Schmidt number $r \geq \max\{d, c d^{-3/5} [\ln(1/\varepsilon)]^{1/10}\}$.

At fixed dimension, these results force double-logarithmic growth of $\log_2 K$, or of the number of initial resource qubits in LOSCC, for almost every target. Section 8 gives the same order within the Haar-null family of diagonal gates by following one scalar invariant; this is an application of the method rather than a consequence of the whole-group measure bound.

1.2 Proof idea and reading guide

The exact argument starts from a frozen dilation: if a protocol implements a unitary U exactly, its discarded environment is independent of the input. After compression, encoder and decoder isometries satisfy $\text{Enc} = \text{Dec}U$, and their infinitesimal generators are local. Differentiation confines the target to a local-unitary orbit; finitely many semialgebraic components then give finitely many orbits per architecture.

For an approximate protocol, we first freeze the environment up to error $O(\sqrt{\varepsilon})$, with Schmidt rank controlled by K (Section 3). We compress the uncharged registers and complete the reversed local decoders to isometries, obtaining a common finite-dimensional witness. Its overlap $\hat{T} = \text{Dec}^\dagger \text{Enc}$ is close to the target and satisfies

$$\hat{T} = -b\hat{T} + \hat{T}a + \Xi, \quad \|\Xi\|_F \leq \sqrt{h}(\|\dot{\text{Enc}}\|_{\text{op}} + \|\dot{\text{Dec}}\|_{\text{op}}),$$

where $h = d^2 - \|\hat{T}\|_F^2$ is the leakage (Section 4). The first two terms are local-unitary directions; for measurements the output generator is diagonal instead.

Section 5 bounds the transverse volume of these low-leakage witness images, using quantitative approximate definable choice [LRT25]. Comparing $O(d^2 K^2)$ parameters with $\Theta(d^4)$ transverse target directions proves Theorem A. Section 6 gives the smaller witness families for Theorem B. Section 7 then compresses even continuous classical transcripts to finite ones without decreasing the score, and applies the charged-message bounds to prove Theorem C.

Readers interested in the diagonal-gate application can proceed from the witness construction to Section 8, which uses the scalar image-volume lemma rather than the tube estimate. Appendices A and B contain the geometric details and elementary rank floors.

Section 9 summarizes the Lean coverage, its three external hypotheses and its limits; the declaration table is in the [repository documentation](#). Appendix C gives an effective bound for the named gate $C_1 = \text{diag}(1, 1, 1, e^i)$, namely $\log_2 K_\varepsilon(C_1) \geq \frac{1}{2} \log_2 \ln \ln(1/\varepsilon) - O(1)$. That arithmetic argument is independent of the geometric precision bounds developed here.

1.3 Related work

Upper bounds. Every bipartite unitary can be approximated in NLQC [Vai03, BCF⁺14]; the port-based construction of Beigi and König [BK11] with the bounds of [CLM⁺21] uses $O(n 2^{4n}/\varepsilon)$ ebits for diamond error ε , and $O(\varepsilon^{-1/2})$ ebits at fixed n . For controlled unitaries and two-qubit unitaries, $O(\log(1/\varepsilon))$ ebits suffice [Yu11, YGC12, GC20]. For comparison, at fixed n our lower bounds on $\log_2 K$, and on the number of resource qubits when only classical messages are sent, are of order $\log \log(1/\varepsilon)$.

Lower bounds. Most lower bounds on NLQC concern the growth of the resources with the input size at a fixed error, often motivated by quantum position verification [BK11, TFKW13, BCS22, ACCM25]. Rank and entanglement lower bounds are known for f -routing and unitary implementation [ACM25, CM26, Bog26]; for instance, Bogner proves an $\Omega(\log n)$ entanglement-of-formation lower bound for inner-product f -routing at constant error [Bog26]. Using the geometry of Banach spaces, exponential resource lower bounds have been proved for restricted classes of strategies [JKPPG22], recently for strategies whose first round has low degree as a function of the classical inputs [MCPPG26]. The latter hypothesis concerns the whole family of first-round operations across inputs, not a single target; it gives family-wise hardness rather than a fixed target that is hard for every protocol. Our bounds are complementary: they concern the dependence on the precision at a given input size, uniformly in n , and for a single target they apply to every protocol. At constant error our methods give only the elementary floors, of order d^2 for universal approximation of unitaries and d^3 for measurements.

Methods. Approximate compression of infinitesimal generators underlies the approximate Eastin–Knill theorems [FNA⁺20]. Dimension counting for the semialgebraic set of unitaries reachable by a fixed circuit architecture gives linear growth of exact circuit complexity [HFK⁺22]; robust versions use different tools [Haf23]. Our tube estimate is a dimension-uniform variant of quantitative Sard theorems in tame geometry [YC04], based on the approximate definable choice of Lerario, Rizzi and Tiberio [LRT25].

2 Setting

We use the protocol model and the notation of [FGMS26], which we briefly recall, and then introduce the error measures and the resource measure used in these notes.

2.1 Protocols and exact rigidity

Alice and Bob receive the input registers A and B , with $H_A \cong H_B \cong \mathbb{C}^d$ and $d = 2^n$. They share a resource state on $R_A R_B$, apply local encoding isometries

$$V_A : AR_A \rightarrow K_A M_{A_{\text{pre}}}, \quad V_B : BR_B \rightarrow K_B M_{B_{\text{pre}}},$$

exchange the message registers with the exchange unitary X_{ex} , which sends $M_{A_{\text{pre}}}$ to $M_{B_{\text{post}}}$ and $M_{B_{\text{pre}}}$ to $M_{A_{\text{post}}}$, and apply local decoding isometries

$$W_A : K_A M_{A_{\text{post}}} \rightarrow A' G_A, \quad W_B : K_B M_{B_{\text{post}}} \rightarrow B' G_B.$$

Auxiliary registers of the Stinespring dilations are included in the kept registers K_A, K_B and the garbage registers G_A, G_B , as in [FGMS26]. The message dimensions are

$$m_A = \dim M_{A_{\text{pre}}} = \dim M_{B_{\text{post}}}, \quad m_B = \dim M_{B_{\text{pre}}} = \dim M_{A_{\text{post}}}.$$

For a pure resource $|\eta\rangle$, the protocol is the isometry

$$F = (W_A \otimes W_B) X_{\text{ex}} (V_A \otimes V_B) E_\eta : AB \rightarrow A' G_A B' G_B, \quad (1)$$

where E_v is the insertion map $E_v \psi = \psi \otimes v$, and it implements the channel $\mathcal{N}_F(\rho) = \text{Tr}_{G_A G_B}(F \rho F^\dagger)$.

Mixed resources. In [FGMS26] the resource is a pure state. For exact implementation this is no restriction, since a purifying register can be given to one party, but purification can increase the Schmidt rank, and the Schmidt rank is one of the resources we charge. We therefore allow a mixed resource σ on $R_A R_B$. Its *Schmidt number* is the least r such that σ is a convex combination of pure states of Schmidt rank at most r . For such a decomposition $\sigma = \sum_j p_j |\eta_j\rangle\langle\eta_j|$, the protocol implements the average $\sum_j p_j \mathcal{N}_{F_j}$ of the channels of the pure protocols F_j with resource $|\eta_j\rangle$. The Choi infidelity and the failure probability of Section 2.2, which our lower bounds use, are affine in the implemented channel, so some pure component has at least the average score; this is how we reduce to pure resources. Shared classical randomness is part of the resource: a seed j known to both parties, together with an entangled state $|\eta_j\rangle$, is the classically flagged mixture $\sum_j p_j |jj\rangle\langle jj| \otimes |\eta_j\rangle\langle\eta_j|$, whose Schmidt number is at most $\max_j \text{SR}(\eta_j)$, and selecting a pure component fixes the seed. Purifying the seed coherently would instead increase the Schmidt rank. Local instruments with classical outcomes and local randomness are included by coherent dilation: outcomes are recorded coherently, later operations are controlled on them, and copies of dephased labels are placed in the garbage registers.

Measurements. For a rank-one projective measurement (PVM) $\Phi = (P_1, \dots, P_{d^2})$ with $P_i = |\varphi_i\rangle\langle\varphi_i|$, the output registers A', B' are label registers of dimension d^2 , read out in the basis $|1\rangle, \dots, |d^2\rangle$, and both parties must output the correct label. The *basis matrix* $M = \sum_i |\varphi_i\rangle\langle i|$ is determined by Φ up to $M \mapsto M\Delta$ with Δ diagonal unitary.

Local orbits. Write $\mathcal{L} = \{L_A \otimes L_B : L_A, L_B \in \text{U}(d)\}$ for the local unitary group. As in [FGMS26], the orbit of a unitary and the orbit of a basis matrix are

$$\mathfrak{o}(U) = \{LUR : L, R \in \mathcal{L}\}, \quad \mathfrak{o}_m(M) = \{LM\Delta : L \in \mathcal{L}, \Delta \in \text{U}(d^2) \text{ diagonal}\},$$

and the corresponding infinitesimal generators are the local anti-Hermitian operators

$$\mathfrak{u}_{\text{loc}} = \{a_A \otimes I + I \otimes a_B : a_A^\dagger = -a_A, a_B^\dagger = -a_B\}$$

and the diagonal anti-Hermitian matrices $\mathfrak{u}_{\text{diag}}$.

Exact rigidity. The exact proof starts from a frozen dilation. If F implements U exactly, then $F = E_\gamma U$ for a unit vector $|\gamma\rangle \in G_A G_B$, by the rigidity lemma of [FGMS26]. If F implements Φ exactly, then $F = C_\gamma M^\dagger$, where $C_\gamma|i\rangle = |i\rangle_{A'}|i\rangle_{B'}|\gamma_i\rangle$ is the duplicated-label map of unit vectors $|\gamma_1\rangle, \dots, |\gamma_{d^2}\rangle \in G_A G_B$, by its measurement version [FGMS26]. To treat both cases at once we write

$$(\Gamma, T) = (E_\gamma, U) \text{ for unitaries,} \quad (\Gamma, T) = (C_\gamma, M^\dagger) \text{ for measurements,} \quad (2)$$

so that exactness reads $F = \Gamma T$ with $T \in \mathbf{U}(d^2)$. With

$$\text{Enc} = X_{\text{ex}}(V_A \otimes V_B)E_\eta, \quad \text{Dec} = (W_A \otimes W_B)^\dagger \Gamma,$$

exactness gives $\text{Enc} = \text{Dec}T$, and along a differentiable path of exact strategies [FGMS26]

$$\dot{T} = -bT + Ta, \quad a = \text{Enc}^\dagger \dot{\text{Enc}} \in \mathfrak{u}_{\text{loc}}, \quad b = \text{Dec}^\dagger \dot{\text{Dec}} \in \begin{cases} \mathfrak{u}_{\text{loc}} & \text{for unitaries,} \\ \mathfrak{u}_{\text{diag}} & \text{for measurements.} \end{cases} \quad (3)$$

For measurements this is the adjoint of the identity $\dot{M} = -aM + Mb$ of [FGMS26].

For an arbitrary complex $d^2 \times d^2$ matrix Y we call

$$\mathcal{T}_{\text{u}}(Y) = \{-bY + Ya : a, b \in \mathfrak{u}_{\text{loc}}\}, \quad \mathcal{T}_{\text{m}}(Y) = \{-bY + Ya : a \in \mathfrak{u}_{\text{loc}}, b \in \mathfrak{u}_{\text{diag}}\} \quad (4)$$

the *allowed directions* at Y . At a unitary $Y = U$, $\mathcal{T}_{\text{u}}(U)$ is the tangent space of $\mathfrak{o}(U)$; at $Y = M^\dagger$, $\mathcal{T}_{\text{m}}(M^\dagger)$ is the tangent space of $\{M'^\dagger : M' \in \mathfrak{o}_{\text{m}}(M)\}$. The real dimension of $\mathfrak{u}_{\text{loc}}$ is $2d^2 - 1$: the map $(a_A, a_B) \mapsto a_A \otimes I + I \otimes a_B$ on pairs of anti-Hermitian matrices, a space of real dimension $2d^2$, has the one-dimensional kernel spanned by $(iI, -iI)$. At every Y , the kernel of the map $(a, b) \mapsto -bY + Ya$, on $\mathfrak{u}_{\text{loc}} \oplus \mathfrak{u}_{\text{loc}}$ and on $\mathfrak{u}_{\text{loc}} \oplus \mathfrak{u}_{\text{diag}}$ alike, contains the line spanned by (iI, iI) . Hence, at every matrix Y ,

$$\dim_{\mathbb{R}} \mathcal{T}_{\text{u}}(Y) \leq \ell_{\text{u}} := 4d^2 - 3, \quad \dim_{\mathbb{R}} \mathcal{T}_{\text{m}}(Y) \leq \ell_{\text{m}} := 3d^2 - 2, \quad (5)$$

in agreement with the orbit dimensions counted in [FGMS26]. The group $\mathbf{U}(d^2)$ has real dimension $N := d^4$, and we write

$$\kappa_{\text{u}} := N - \ell_{\text{u}} = (d^2 - 1)(d^2 - 3), \quad \kappa_{\text{m}} := N - \ell_{\text{m}} = (d^2 - 1)(d^2 - 2) \quad (6)$$

for the corresponding codimensions; since ℓ bounds the dimension of every orbit, κ is a lower bound on the codimension of every orbit. Both are at least $\frac{3}{16}d^4$ for every $d \geq 2$. When a statement applies to both target classes, we drop the subscripts and write $\mathcal{T}$, ℓ and κ .

2.2 Error measures

Let $\bar{A} \cong A$ and $\bar{B} \cong B$ be reference registers and

$$|\Omega\rangle = \frac{1}{d} \sum_{j=1}^{d^2} |j\rangle_{AB} |j\rangle_{\bar{A}\bar{B}}$$

the maximally entangled state. For an operator X with domain AB we write $|X\rangle\rangle = (X \otimes I_{\bar{A}\bar{B}})|\Omega\rangle$ for its normalized Choi vector. Then

$$\|X\|_{\text{n}} := \frac{1}{d} \|X\|_F = \||X\rangle\rangle\| \quad (7)$$

is the *normalized Frobenius norm*, which gives unitaries norm one. For two isometries F, F' with the same domain and codomain, $\|F - F'\|_{\text{n}}^2 = 2 - 2d^{-2} \text{Re Tr}(F'^\dagger F)$.

Unitaries. The Choi state of a channel $\mathcal{N}$ is $\rho_{\mathcal{N}} = (\mathcal{N} \otimes \text{id})(|\Omega\rangle\langle\Omega|)$. The *score* of $\mathcal{N}$ for a target unitary U and its *Choi infidelity* are

$$q_U(\mathcal{N}) = \langle\langle U | \rho_{\mathcal{N}} | U \rangle\rangle, \quad \delta_{\text{Choi}}(\mathcal{N}, U) = 1 - q_U(\mathcal{N}). \quad (8)$$

Thus q_U is the fidelity of the Choi state with the pure target Choi state, and δ_{Choi} is the entanglement infidelity. For the channel $\mathcal{N}_F$ of a pure protocol, $q_U(\mathcal{N}_F) = \|(\langle\langle U | \otimes I_{G_A G_B}) | F \rangle\rangle\|^2$. The normalized diamond distance $\delta_{\diamond}(\mathcal{N}, U) = \frac{1}{2} \|\mathcal{N} - \mathcal{U}\|_{\diamond}$, with $\mathcal{U}(\rho) = U\rho U^\dagger$, satisfies

$$\delta_{\text{Choi}}(\mathcal{N}, U) \leq \delta_{\diamond}(\mathcal{N}, U), \quad (9)$$

because $1 - \langle\langle U | \rho_{\mathcal{N}} | U \rangle\rangle$ is at most the trace distance between $\rho_{\mathcal{N}}$ and the pure state $\rho_{\mathcal{U}} = |U\rangle\langle U|$. All our lower bounds only use $\delta_{\text{Choi}} \leq \varepsilon$, and therefore also hold under $\delta_{\diamond} \leq \varepsilon$.

Measurements. Let $p_{\rho}(a, b)$ be the probability that Alice outputs a and Bob outputs b on input ρ . The score of a protocol for the PVM Φ and its *failure probability* are

$$q_{\Phi} = \frac{1}{d^2} \sum_{i=1}^{d^2} p_{\varphi_i}(i, i), \quad \delta_{\Phi} = 1 - q_{\Phi}. \quad (10)$$

This is the probability that both parties output the correct label when the input is a uniformly random basis vector; it does not depend on the phases of the φ_i . The worst-case total-variation error $\sup_{\rho} \frac{1}{2} \sum_{a,b} |p_{\rho}(a, b) - \delta_{ab} \text{Tr}(P_a \rho)|$ is at least δ_{Φ} , as one sees by taking $\rho = |\varphi_i\rangle\langle\varphi_i|$ and averaging over i .

2.3 The footprint

Definition 2.1 (Footprint). *A protocol has footprint at most K if its resource has Schmidt number r and its messages have dimensions m_A, m_B with*

$$r m_A m_B \leq K.$$

The footprint charges the full dimension of each message, whether the message is quantum or classical; a classical message with m possible values is a message register of dimension m . The kept registers K_A, K_B (not to be confused with the footprint K), the garbage registers G_A, G_B and all auxiliary registers are free; the proofs compress them. For exact protocols with finite-dimensional resources, classical and quantum messages are equivalent, since quantum messages can be teleported with additional finite entanglement [FGMS26]. This conversion changes the footprint, so quantitatively the two models differ; Section 7 treats the model in which classical messages are free.

For $K \geq 1$ and $\varepsilon \geq 0$, let $\text{Reach}_{K,\varepsilon}^u \subseteq \text{U}(d^2)$ be the set of unitaries U for which some protocol of footprint at most K has $\delta_{\text{Choi}} \leq \varepsilon$, and let $\text{Reach}_{K,\varepsilon}^m \subseteq \text{U}(d^2)$ be the set of matrices $M^\dagger$, where M is a basis matrix of a PVM that some protocol of footprint at most K implements with $\delta_{\Phi} \leq \varepsilon$. The second set is invariant under $M^\dagger \mapsto \Delta M^\dagger$ for diagonal unitaries Δ , so its Haar measure equals the invariant measure of the corresponding set of PVMs in $\text{U}(d^2)/\text{U}(1)^{d^2}$. When a statement applies to both target classes we drop the superscripts and write $\text{Reach}_{K,\varepsilon}$. For a target T we write

$$K_{\varepsilon}(T) = \min\{K : T \in \text{Reach}_{K,\varepsilon}\}$$

for its least footprint at error ε , with the appropriate reachable set.

2.4 Qualitative divergence

The exact results of [FGMS26], together with compactness, already imply that the footprint must diverge for targets without an exact protocol.

Proposition 2.2 (Qualitative divergence). *Fix $K \geq 1$ and a target, either a unitary U or a PVM Φ . Among protocols of footprint at most K , the least error $\varepsilon_K(T)$ is attained, and $\varepsilon_K(T) = 0$ if and only if the target has an exact protocol of footprint at most K . The sets $\text{Reach}_{K,\varepsilon}$ are compact. Consequently, if the target has no exact protocol with finite-dimensional resources, then $K_\varepsilon(T) \rightarrow \infty$ as $\varepsilon \rightarrow 0$.*

Proof. The error is affine in the resource state, so we may restrict to pure resources of Schmidt rank at most r , and hence to $R_A \cong R_B \cong \mathbb{C}^r$. By Lemma 4.1 below, the kept registers can be taken of dimension $k_X \leq drm_X$, so the decoders act on spaces of dimension at most dK . The error only depends on the channel $\text{Tr}_{G_X}(W_X \cdot W_X^\dagger)$ implemented by each decoder, where $X \in \{A, B\}$, and a channel from a space of dimension q to one of dimension q' has a Stinespring isometry with environment dimension at most qq' . Thus finitely many choices of register dimensions suffice, and for each of them the protocols form a compact set, a product of unit spheres and sets of isometries. The error is a continuous function of the target and the protocol, so its minimum is attained, and the set of targets with error at most ε is a finite union of projections of compact sets, hence compact.

If a unitary protocol has error zero, its Choi state has fidelity one with the pure state $|U\rangle\rangle\langle\langle U|$, so the two are equal and the channel is $\rho \mapsto U\rho U^\dagger$. If a measurement protocol has error zero, let E_{ab} be the effect on the input that leads to the outputs (a, b) . Then $\langle\varphi_i|E_{ii}|\varphi_i\rangle = 1$ for all i , so positivity and $\sum_{a,b} E_{ab} = I$ give $E_{ab}|\varphi_i\rangle = 0$ for $(a, b) \neq (i, i)$; hence $E_{ab} = 0$ for $a \neq b$ and $E_{ii} = |\varphi_i\rangle\langle\varphi_i|$, which is exactness. Finally, if no exact protocol exists, then $\varepsilon_K(T) > 0$ for every K , and $K_\varepsilon(T) > K$ whenever $\varepsilon < \varepsilon_K(T)$. $\square$

The proposition gives no rate, since nothing controls how fast $\varepsilon_K(T)$ decreases with K , and it is not uniform over targets. The remaining sections supply rates.

3 Approximately frozen dilations

The exact proof starts from the observation that the environment of an exact protocol ends in a fixed state. For an accurate protocol, we now construct a nearby frozen dilation whose environment state has Schmidt rank bounded by the footprint. This bound is what later makes the number of parameters independent of the uncharged garbage registers.

3.1 Schmidt rank across the laboratory cut

The Choi vector $|F\rangle\rangle$ of a protocol lies in $A'G_AB'G_B\bar{A}\bar{B}$. The *laboratory cut* separates Alice's systems $A'G_A\bar{A}$ from Bob's systems $B'G_B\bar{B}$; for vectors in G_AG_B we use the cut between G_A and G_B . For a vector χ in a bipartite space, not necessarily normalized, let $\text{SR}(\chi)$ be its Schmidt rank, let $\lambda_1 \geq \lambda_2 \geq \dots$ be its squared Schmidt coefficients (padded with zeros), and let

$$\Lambda_k(\chi) = \sum_{j=1}^k \lambda_j = \max\{|\langle\chi'|\chi\rangle|^2 : \|\chi'\| = 1, \text{SR}(\chi') \leq k\}, \quad \Lambda_0(\chi) = 0, \quad (11)$$

be its *top- k Schmidt mass*; thus $\Lambda_k(\chi) \leq \|\chi\|^2$, with equality for $k \geq \text{SR}(\chi)$. The variational formula is the Eckart–Young theorem for the best rank- k approximation of the coefficient matrix of χ in Frobenius norm [Bha97]. For unit vectors χ and χ' , each in a bipartite space, the product $\chi \otimes \chi'$ has the squared Schmidt coefficients $\lambda_i(\chi)\lambda_j(\chi')$, and

$$\Lambda_k(\chi \otimes \chi') \leq \min\{\Lambda_k(\chi), \Lambda_k(\chi')\}. \quad (12)$$

Indeed, any k pairs (i, j) involve at most k distinct indices j , and grouping the pairs by j bounds their mass by $\Lambda_k(\chi')$; the other bound is symmetric.

Lemma 3.1 (Schmidt rank of a protocol). *If the resource $|\eta\rangle$ has Schmidt rank r , then $\text{SR}(|F\rangle) \leq r m_A m_B$ across the laboratory cut.*

Proof. Before the protocol, the input registers are maximally entangled with the references, A with $\bar{A}$ and B with $\bar{B}$, and the state of $A\bar{A}R_A$ versus $B\bar{B}R_B$ has Schmidt rank r . The encoders are local isometries and do not change the Schmidt rank. Moving a register of dimension m from one laboratory to the other multiplies the Schmidt rank by at most m : expanding that register in a basis writes every Schmidt term as a sum of m product terms across the new cut. The exchange moves $M_{A_{\text{pre}}}$ and $M_{B_{\text{pre}}}$, and the decoders are again local. $\square$

3.2 Approximately frozen dilations

Lemma 3.2 (Approximately frozen dilations). *Let a protocol have footprint at most K , and let $0 \leq \varepsilon < 1$.*

- (i) *If it implements U with $\delta_{\text{Choi}} \leq \varepsilon$, then some pure component of its resource gives a protocol isometry F and a unit vector $|\gamma\rangle \in G_A G_B$ with*

$$\text{SR}(\gamma) \leq K, \quad \|F - E_\gamma U\|_{\text{n}} \leq \sqrt{2\varepsilon}. \quad (13)$$

- (ii) *If it implements Φ with $\delta_\Phi \leq \varepsilon$, then some pure component gives a protocol isometry F and unit vectors $|\gamma_1\rangle, \dots, |\gamma_{d^2}\rangle \in G_A G_B$ with*

$$\sum_{i=1}^{d^2} \text{SR}(\gamma_i) \leq K + d^2, \quad \|F - C_\gamma M^\dagger\|_{\text{n}} \leq 2\sqrt{\varepsilon}. \quad (14)$$

In the notation (2), both parts say that $\|F - \Gamma T\|_{\text{n}} \leq \delta$ with

$$\delta = \sqrt{2\varepsilon} \text{ for unitaries,} \quad \delta = 2\sqrt{\varepsilon} \text{ for measurements.} \quad (15)$$

At $\varepsilon = 0$ the lemma reduces to the rigidity lemma of [FGMS26] and its measurement version, with the additional information that the environment can be taken of Schmidt rank at most K (respectively $K + d^2$).

Proof. Both scores are affine in the resource state, so a decomposition witnessing Schmidt number at most r contains a pure component with score at least $1 - \varepsilon$. Fix it and let $v = |F\rangle$. By Lemma 3.1, v is a unit vector with $\text{SR}(v) \leq K$.

(i) Project onto the target Choi vector: let $\xi = (\langle U | \otimes I_{G_A G_B})v \in G_A G_B$. The score is $p := \|\xi\|^2 = q_U \geq 1 - \varepsilon > 0$, and we put $\gamma' = \xi/\sqrt{p}$. The unit vector $|U\rangle \otimes \gamma'$ has overlap $\sqrt{p}$ with v , so (11) with $\text{SR}(v) \leq K$ and (12) give

$$p \leq \Lambda_K(|U\rangle \otimes \gamma') \leq \Lambda_K(\gamma') =: \tau.$$

Let γ be the normalized truncation of γ' to its K largest Schmidt terms, so that $\text{SR}(\gamma) \leq K$ and $\langle \gamma | \gamma' \rangle = \sqrt{\tau}$. The Choi vector of $E_\gamma U$ is $|U\rangle \otimes \gamma$, and

$$(\langle U | \otimes \langle \gamma |)v = \langle \gamma | \xi \rangle = \sqrt{p\tau} \geq p.$$

Both v and $|U\rangle \otimes \gamma$ are unit vectors, so $\|F - E_\gamma U\|_{\text{n}}^2 \leq 2 - 2p \leq 2\varepsilon$.

(ii) Write $|\Omega\rangle = d^{-1} \sum_i |\varphi_i\rangle |\bar{\varphi}_i\rangle$, with $\bar{\varphi}_i$ the complex conjugate of φ_i in the reference. Let Π be the projector onto the mutually orthogonal subspaces $|i\rangle_{A'} |i\rangle_{B'} |\bar{\varphi}_i\rangle_{\bar{A}\bar{B}} \otimes G_A G_B$. Then

$$v_0 := \Pi v = \frac{1}{d} \sum_{i=1}^d |i\rangle_{A'} |i\rangle_{B'} |\bar{\varphi}_i\rangle |\tilde{\gamma}_i\rangle, \quad \tilde{\gamma}_i = (\langle i|_{A'} \langle i|_{B'} \otimes I) F |\varphi_i\rangle, \quad (16)$$

with $\|\tilde{\gamma}_i\| \leq 1$, and the score is $p := \|v_0\|^2 = d^{-2} \sum_i \|\tilde{\gamma}_i\|^2 = q_\Phi \geq 1 - \varepsilon$. Since $\langle v | v_0 \rangle = p$ and v is a unit vector of Schmidt rank at most K , (11) applied to the unnormalized vector v_0 gives $\Lambda_K(v_0) \geq p^2$.

The two copies of the label make the Schmidt decomposition of v_0 block diagonal: the terms with different i are orthogonal on both sides of the laboratory cut. If ζ_{ik}^2 are the squared Schmidt coefficients of $\tilde{\gamma}_i$ and λ_{ij} those of the unit vector φ_i , the squared Schmidt coefficients of v_0 are

$$d^{-2} \zeta_{ik}^2 \lambda_{ij}. \quad (17)$$

Select K triples (i, j, k) that carry the top- K mass of v_0 , and let $\hat{\gamma}_i$ be the part of the Schmidt decomposition of $\tilde{\gamma}_i$ consisting of the terms k that occur in a selected triple. Since $\sum_j \lambda_{ij} = 1$, these vectors satisfy

$$\sum_i \text{SR}(\hat{\gamma}_i) \leq K, \quad d^{-2} \sum_i \|\hat{\gamma}_i\|^2 \geq \Lambda_K(v_0) \geq p^2.$$

Put $t_i = \|\hat{\gamma}_i\| \leq 1$. If $t_i > 0$, let $\gamma_i = \hat{\gamma}_i/t_i$; orthogonality of the Schmidt terms gives $\langle \gamma_i | \tilde{\gamma}_i \rangle = t_i$. If $t_i = 0$, let γ_i be a unit product vector, chosen as a Schmidt term of $\tilde{\gamma}_i$ with a phase that makes $\langle \gamma_i | \tilde{\gamma}_i \rangle \geq 0$ if $\tilde{\gamma}_i \neq 0$. This adds at most one Schmidt term per outcome, so $\sum_i \text{SR}(\gamma_i) \leq K + d^2$. The Choi vector of $C_\gamma M^\dagger$ is $d^{-1} \sum_i |i\rangle |i\rangle |\bar{\varphi}_i\rangle |\gamma_i\rangle$, a unit vector, and

$$\text{Re} \langle |C_\gamma M^\dagger\rangle, v \rangle = \frac{1}{d^2} \sum_i \text{Re} \langle \gamma_i | \tilde{\gamma}_i \rangle \geq \frac{1}{d^2} \sum_i t_i \geq \frac{1}{d^2} \sum_i t_i^2 \geq p^2.$$

Hence $\|F - C_\gamma M^\dagger\|_{\text{n}}^2 \leq 2 - 2p^2 \leq 4\varepsilon$. $\square$

Remark 3.3 (Which error the proof uses). The proof only uses the score of the selected pure component. That component need not satisfy the operational error bound of the mixed protocol, for instance in diamond distance, but its score suffices for every later step. This is why all our lower bounds are stated for the scores, and why they transfer to the free classical model of Section 7, where the compression only preserves the score.

4 The perturbed derivative identity

We now construct, for each accurate protocol, two isometries with a common codomain whose overlap is close to the target, and prove that the derivative of this overlap consists of allowed directions plus a small leakage term. The construction uses only coordinate spaces whose dimensions are bounded in terms of K and d .

4.1 A common finite-dimensional witness

First we remove the dimensions of the kept registers that a protocol never uses. For a pure resource of Schmidt rank r we restrict R_A and R_B to the supports of the reduced states, of dimension r .

Lemma 4.1 (Reachable-space compression). *The kept registers can be replaced by subspaces $\tilde{K}_A \subseteq K_A$ and $\tilde{K}_B \subseteq K_B$ with $\dim \tilde{K}_X \leq dr m_X$ for $X \in \{A, B\}$, without changing F . The decoder inputs $Q_A := \tilde{K}_A M_{A_{\text{post}}}$ and $Q_B := \tilde{K}_B M_{B_{\text{post}}}$ then have dimension at most $dr m_A m_B \leq dK$.*

Proof. The domain of V_A has dimension at most dr . Let $\widetilde{K}_A$ be the span of the vectors $(I_{K_A} \otimes \langle \mu |) V_A | \psi \rangle$, where $|\psi\rangle$ runs over a basis of the domain and $|\mu\rangle$ over a basis of $M_{A_{\text{pre}}}$; it has dimension at most drm_A , and the range of V_A lies in $\widetilde{K}_A M_{A_{\text{pre}}}$. After the exchange, Alice's decoder only acts on $\widetilde{K}_A M_{A_{\text{post}}}$, of dimension at most $drm_A m_B$. The same holds for Bob. $\square$

This is the compression argument of [FGMS26], which there shows that only the Schmidt rank and the message dimensions matter for the finite-orbit theorem. We then pad the dimensions to fixed values: the resource registers to $\mathbb{C}^r$ and the kept registers to dimension drm_X , so that the decoder inputs Q_X have dimension $drm_A m_B \leq dK$. An encoder extends isometrically to added resource directions by completing its orthonormal columns, which is possible because its padded codomain has dimension $drm_X^2 \geq dr$, and a decoder extends to added input directions after enlarging its garbage register. Since the resource has no amplitude on the added directions, these extensions do not change F .

Support coordinates. Next we replace the uncharged garbage registers by coordinate spaces whose dimension is bounded by the footprint, using the environment of Lemma 3.2. Write Γ_{ph} for the physical frozen map of that lemma, with environment vector γ_{ph} or vectors $\gamma_{i,\text{ph}}$ in $G_A G_B$; the unsubscripted Γ and γ will denote their coordinate versions.

Unitaries. Choose isometries $j_X : \mathbb{C}^K \rightarrow G_X$ whose ranges contain the local Schmidt supports of γ_{ph} , enlarging G_X if necessary. Put

$$\widehat{G}_X = X' \otimes \mathbb{C}^K, \quad \iota_X = I_{X'} \otimes j_X : \widehat{G}_X \rightarrow X' G_X, \quad \gamma = (j_A^\dagger \otimes j_B^\dagger) \gamma_{\text{ph}}, \quad \Gamma = E_\gamma.$$

Measurements. Let $s_i = \text{SR}(\gamma_{i,\text{ph}}) \geq 1$ and $S = \sum_i s_i \leq K + d^2$. Choose isometries $j_{X,i} : \mathbb{C}^{s_i} \rightarrow G_X$ whose ranges contain the local Schmidt supports of $\gamma_{i,\text{ph}}$, and put

$$\widehat{G}_X = \bigoplus_{i=1}^{d^2} |i\rangle \otimes \mathbb{C}^{s_i}, \quad \iota_X(|i\rangle \otimes w) = |i\rangle_{X'} \otimes j_{X,i} w, \quad \gamma_i = (j_{A,i}^\dagger \otimes j_{B,i}^\dagger) \gamma_{i,\text{ph}},$$

and let $\Gamma = C_\gamma : \mathbb{C}^{d^2} \rightarrow \widehat{G}_A \otimes \widehat{G}_B$ map $|i\rangle$ to its two labels in $\widehat{G}_A$ and $\widehat{G}_B$ and the vector γ_i . The map ι_X is an isometry because distinct labels are orthogonal, even though the physical supports of different $\gamma_{i,\text{ph}}$ need not be. In both cases $\widehat{G}_X$ replaces the output and garbage space $X' G_X$; it has dimension dK or S , and

$$(\iota_A \otimes \iota_B) \Gamma = \Gamma_{\text{ph}}. \tag{18}$$

Completing the reversed decoders. Running the decoder backwards on these supports gives the contraction $W_X^{\text{rev}} := W_X^\dagger \iota_X : \widehat{G}_X \rightarrow Q_X$. It fails to be an isometry exactly to the extent that the environment support leaves the range of the decoder, which can happen for an approximate protocol. We complete it to an isometry by adding a local block that records this missing amplitude:

$$J_X = \begin{pmatrix} W_X^{\text{rev}} \\ Z_X \end{pmatrix} : \widehat{G}_X \rightarrow Q_X \oplus \widehat{G}_X, \quad Z_X = (I - W_X^{\text{rev} \dagger} W_X^{\text{rev}})^{1/2}, \tag{19}$$

where the square root is the positive semidefinite one; then $J_X^\dagger J_X = W_X^{\text{rev} \dagger} W_X^{\text{rev}} + Z_X^2 = I$. In an exact protocol with minimal support coordinates, W_X^{rev} is already an isometry and $Z_X = 0$.

The two isometries. Let $\mathcal{Q} = (Q_A \oplus \hat{G}_A) \otimes (Q_B \oplus \hat{G}_B)$, which contains $Q_A \otimes Q_B$. The encoder isometry $\text{Enc} = X_{\text{ex}}(V_A \otimes V_B)E_\eta$ of [FGMS26] maps AB into $Q_A \otimes Q_B \subseteq \mathcal{Q}$. We define the decoder isometry and the overlap by

$$\text{Dec} = (J_A \otimes J_B)\Gamma : \mathbb{C}^{d^2} \rightarrow \mathcal{Q}, \quad \hat{T} = \text{Dec}^\dagger \text{Enc}. \quad (20)$$

This Dec replaces the decoder isometry $(W_A \otimes W_B)^\dagger \Gamma$ of the exact proof, with which it agrees when $Z_A = Z_B = 0$. Only the blocks W_X^{rev} meet the range of Enc , so by (18)

$$\hat{T} = \Gamma^\dagger (W_A^{\text{rev}\dagger} \otimes W_B^{\text{rev}\dagger}) \text{Enc} = \Gamma^\dagger (\iota_A^\dagger \otimes \iota_B^\dagger) (W_A \otimes W_B) \text{Enc} = \Gamma_{\text{ph}}^\dagger F. \quad (21)$$

For two isometries, the squared normalized distance is determined by their overlap. Hence, with δ as in (15),

$$\|\text{Enc} - \text{Dec}\hat{T}\|_n^2 = 2 - \frac{2}{d^2} \text{Re Tr}(T^\dagger \hat{T}) = \|F - \Gamma_{\text{ph}} T\|_n^2 \leq \delta^2. \quad (22)$$

Leakage. The *leakage residuals* and the *leakage* are

$$\begin{aligned} Y_{\text{Enc}} &= \text{Enc} - \text{Dec}\hat{T} = (I - \text{Dec}\text{Dec}^\dagger)\text{Enc}, & Y_{\text{Dec}} &= \text{Dec} - \text{Enc}\hat{T}^\dagger = (I - \text{Enc}\text{Enc}^\dagger)\text{Dec}, \\ h &= \|Y_{\text{Enc}}\|_F^2 = \|Y_{\text{Dec}}\|_F^2 = d^2 - \|\hat{T}\|_F^2. \end{aligned} \quad (23)$$

The equalities follow by expanding the squares and using $\text{Enc}^\dagger \text{Enc} = \text{Dec}^\dagger \text{Dec} = I$. Applying $\text{Dec}^\dagger$ and $I - \text{Dec}\text{Dec}^\dagger$ to $\text{Enc} - \text{Dec}\hat{T}$ gives

$$\|\hat{T} - T\|_n \leq \delta, \quad h \leq d^2 \delta^2. \quad (24)$$

For an exact protocol, $\hat{T} = T$, $h = 0$, and $\text{Enc} = \text{Dec}T$ is the exact factorization of [FGMS26].

4.2 The derivative identity

From now on we forget the protocol that produced the witness and let all blocks vary independently. The variables are the resource $\eta \in \mathbb{C}^r \otimes \mathbb{C}^r$, the encoders V_A, V_B , the completed isometries J_A, J_B , and the environment vector $\gamma \in \mathbb{C}^K \otimes \mathbb{C}^K$ or the vectors $\gamma_i \in \mathbb{C}^{s_i} \otimes \mathbb{C}^{s_i}$. They are subject only to the constraints that vectors are unit vectors and that V_A, V_B, J_A, J_B are isometries; in particular J_X is now an arbitrary isometry $\hat{G}_X \rightarrow Q_X \oplus \hat{G}_X$, not the completion (19) of a particular decoder. For every point of this *constraint space* we define Enc , Dec , $\hat{T}$, Y_{Enc} , Y_{Dec} and h by the formulas above. The constraint space plays the role of the strategy set $\mathcal{Z}_\nu$ of [FGMS26]: the isometry and normalization constraints (Z2) and (Z3) are kept, with J_X in place of W_X , while the target is no longer a coordinate, the constraint (Z4) is dropped, and the overlap $\hat{T}$ plays the role of $U(z)$. Accuracy will be imposed later through the leakage, by the condition $h \leq d^2 \delta^2$. A *path* is a differentiable path in the constraint space; for an isometry block V its velocity satisfies $V^\dagger \dot{V} + \dot{V}^\dagger V = 0$.

Proposition 4.2 (Overlap derivative). *Along every path in the constraint space,*

$$a := \text{Enc}^\dagger \dot{\text{Enc}} \in \mathfrak{u}_{\text{loc}}, \quad b := \text{Dec}^\dagger \dot{\text{Dec}} \in \begin{cases} \mathfrak{u}_{\text{loc}} & \text{for unitaries,} \\ \mathfrak{u}_{\text{diag}} & \text{for measurements,} \end{cases} \quad (25)$$

and

$$\dot{\hat{T}} = -b\hat{T} + \hat{T}a + \Xi, \quad \Xi = \text{Dec}^\dagger Y_{\text{Enc}} + Y_{\text{Dec}}^\dagger \text{Enc}, \quad \|\Xi\|_F \leq \sqrt{h} (\|\dot{\text{Enc}}\|_{\text{op}} + \|\dot{\text{Dec}}\|_{\text{op}}). \quad (26)$$

In particular, the distance of $\dot{\hat{T}}$ from the allowed directions $\mathcal{T}(\hat{T})$ of (4) is at most $\sqrt{h}(\|\dot{\text{Enc}}\|_{\text{op}} + \|\dot{\text{Dec}}\|_{\text{op}})$.

The locality of the generators holds at every point of the constraint space, whether or not the point comes from an accurate protocol; accuracy only enters through the leakage h . For $h = 0$, (26) is the exact identity (3).

Proof. Both Enc and Dec are isometries at every point, so a and b are anti-Hermitian. Differentiate $\hat{T} = \text{Dec}^\dagger \text{Enc}$ and insert $\text{Enc} = \text{Dec} \hat{T} + Y_{\text{Enc}}$ in the first term and $\text{Dec} = \text{Enc} \hat{T}^\dagger + Y_{\text{Dec}}$ in the second:

$$\dot{\hat{T}} = \dot{\text{Dec}}^\dagger (\text{Dec} \hat{T} + Y_{\text{Enc}}) + (\text{Enc} \hat{T}^\dagger + Y_{\text{Dec}})^\dagger \dot{\text{Enc}} = -b \hat{T} + \hat{T} a + \dot{\text{Dec}}^\dagger Y_{\text{Enc}} + Y_{\text{Dec}}^\dagger \dot{\text{Enc}},$$

using $\dot{\text{Dec}}^\dagger \text{Dec} = b^\dagger = -b$. The bound on Ξ follows from $\|Y_1 Y_2\|_F \leq \|Y_1\|_{\text{op}} \|Y_2\|_F$ and $\|Y_1 Y_2\|_F \leq \|Y_1\|_F \|Y_2\|_{\text{op}}$ together with (23).

The input generator is computed exactly as in [FGMS26]: the fixed exchange cancels, and $V_X^\dagger V_X = I$ gives

$$a = E_\eta^\dagger (V_A^\dagger \dot{V}_A \otimes I + I \otimes V_B^\dagger \dot{V}_B) E_\eta + \langle \eta | \dot{\eta} \rangle I.$$

The first two terms are local by the lemma on compression through a shared state in the preliminaries of [FGMS26], and $\langle \eta | \dot{\eta} \rangle$ is imaginary, so $a \in \mathfrak{u}_{\text{loc}}$.

For the output generator, the product rule and $J_X^\dagger J_X = I$ give

$$b = \Gamma^\dagger (J_A^\dagger \dot{J}_A \otimes I + I \otimes J_B^\dagger \dot{J}_B) \Gamma + \Gamma^\dagger \dot{\Gamma}.$$

For unitaries, $\Gamma = E_\gamma$ inserts a state on $\mathbb{C}^K \otimes \mathbb{C}^K$, the operator $J_A^\dagger \dot{J}_A$ acts on Alice's $\hat{G}_A = A' \otimes \mathbb{C}^K$, and the same compression lemma makes the first term local on $A'B'$; moreover $\Gamma^\dagger \dot{\Gamma} = \langle \gamma | \dot{\gamma} \rangle I$ is an imaginary multiple of the identity. For measurements, $\Gamma = C_\gamma$, and each entry $\langle i | C_\gamma^\dagger (J_A^\dagger \dot{J}_A \otimes I) C_\gamma | j \rangle$ contains the overlap $\langle i | j \rangle$ of Bob's two labels, so it vanishes for $i \neq j$; the same holds for Bob's term with Alice's labels. This is the duplicated-label lemma in the measurement appendix of [FGMS26]. Finally $C_\gamma^\dagger \dot{C}_\gamma = \text{diag}(\langle \gamma_1 | \dot{\gamma}_1 \rangle, \dots, \langle \gamma_{d^2} | \dot{\gamma}_{d^2} \rangle)$. Since b is anti-Hermitian, $b \in \mathfrak{u}_{\text{loc}}$ for unitaries and $b \in \mathfrak{u}_{\text{diag}}$ for measurements. $\square$

In the exact proof of [FGMS26], the output compression uses that the range of Γ lies in the range of $W_A \otimes W_B$, which follows from exactness. Here the completed maps J_X are isometries, and no such condition is needed. This is what makes the generator algebras independent of the accuracy of the protocol.

5 Haar volume and generic targets

We now count the coordinates of the witnesses, bound the derivative of the overlap map in these coordinates, and convert the bounds into a volume estimate. This proves Theorem A.

Conventions. All coordinate spaces are real Euclidean spaces: a complex entry contributes its real and imaginary parts, and matrices carry the inner product $\text{Re Tr}(X^\dagger Y)$. Ranks and singular values of derivatives are taken in this real-linear sense. On the target space $\mathbb{C}^{d^2 \times d^2}$ we use the normalized Frobenius norm (7) unless stated otherwise. Constants $c, C, C', \dots$ are universal and may change from line to line; the constant $C_\star$ is fixed once in (27).

5.1 Coordinates and derivative bounds

Fix a factorization (r, m_A, m_B) with $r m_A m_B \leq K$, and for measurements also the ranks $(s_i)_{i \leq d^2}$ of Section 4.1. The blocks of the constraint space have the following numbers of real coordinates. The resource uses $2r^2$. The encoder V_X maps a space of dimension dr to one of dimension $dr m_X^2$, so it uses $2d^2 r^2 m_X^2 \leq 2d^2 K^2$. For unitaries, γ uses $2K^2$, and each J_X

maps $\widehat{G}_X$, of dimension dK , to $Q_X \oplus \widehat{G}_X$, of dimension at most $2dK$, so it uses at most $4d^2K^2$. For measurements, the vectors γ_i use $2\sum_i s_i^2 \leq 2S^2$ and each J_X uses $2(dK + S)S$, where $S \leq K + d^2$. Hence the number of coordinates is at most Cd^2K^2 for unitaries and $Cd^2(K + d^2)^2$ for measurements.

We call the set of points for one choice of these discrete data a *family*; it is the analogue of the strategy set $\mathcal{Z}_\nu$ for one dimension vector ν . There are at most K^3 factorizations, and at most $\binom{K+d^2}{d^2} \leq 2^{K+d^2}$ rank allocations. We pad every family with dummy coordinates to the common number

$$D = \lceil C_\star d^2(K + d^2)^2 \rceil, \quad \text{or} \quad D = \lceil C_\star d^2 K^2 \rceil \text{ if } K \geq d^2/4, \quad (27)$$

with a sufficiently large universal constant $C_\star \geq 16$; the second choice is possible because $K + d^2 \leq 5K$ when $K \geq d^2/4$. The number of families is then at most e^{CD} .

Scaling. We rescale each block to norm one, so that the source has a universal radius. Divide V_X by $\sqrt{dr}$, its Frobenius norm. For unitaries divide J_X by $\sqrt{dK}$ and leave η and γ unchanged. For measurements divide J_X by $\sqrt{S}$, divide the tuple $(\gamma_i)_i$ by d , and leave η unchanged. There are six blocks of norm one, so the rescaled constraint space $\widetilde{\mathcal{Z}}$ lies in the ball of radius $\sqrt{6}$ in $\mathbb{R}^D$. We call

$$\widetilde{\mathcal{Z}}_\delta = \{x \in \widetilde{\mathcal{Z}} : h(x) \leq d^2\delta^2\} \quad (28)$$

the *witness family*; it is compact, being a closed subset of the compact constraint space, since h is continuous. By (24), every target of error at most ε has a witness $x \in \widetilde{\mathcal{Z}}_\delta$ in one of the families, with $\|\widehat{T}(x) - T\|_n \leq \delta$.

Polynomial description. The maps **Enc**, **Dec** and $\widehat{T}$ are polynomials of fixed degree in the entries of the blocks and their complex conjugates; rescaling changes coefficients but not degrees. Each matrix constraint $V^\dagger V = I$ can be written as a single polynomial equation, the sum of the squares of the real and imaginary parts of the entries of $V^\dagger V - I$; the normalizations of the γ_i can be packaged as $\sum_i (\|\gamma_i\|^2 - 1)^2 = 0$. Together with the inequality $h \leq d^2\delta^2$ and the graph equation of $\widehat{T}$, with $2d^4$ output coordinates, this describes the graph of $\widehat{T}$ on $\widetilde{\mathcal{Z}}_\delta$ by a fixed number of polynomial equations and inequalities of fixed degrees in $O(D + d^4)$ real variables.

Speed. Let $\dot{x}$ be the velocity of a path in the rescaled coordinates. By the product rule, $\dot{\mathbf{Enc}}$ is a sum of three terms, containing $\dot{V}_A$, $\dot{V}_B$ and $\dot{\eta}$ respectively, and $\dot{\mathbf{Dec}}$ is a sum of three terms, containing $\dot{J}_A$, $\dot{J}_B$ and $\dot{\Gamma}$; in each term every undifferentiated factor is an isometry and has operator norm one. The derivative of the insertion map E_γ is $E_{\dot{\gamma}}$, of operator norm $\|\dot{\gamma}\|$, and the derivative of C_γ has operator norm $\max_i \|\dot{\gamma}_i\|$. We bound the operator norms of $\dot{V}_X$ and $\dot{J}_X$ by their Frobenius norms, which are $\sqrt{dr}$ and $\sqrt{dK}$ (respectively $\sqrt{S}$) times the coordinate speeds of the rescaled blocks, and the speeds of the vectors by their coordinate speeds, times d for the rescaled tuple $(\gamma_i)_i$. Cauchy–Schwarz over the six terms gives

$$\|\dot{\mathbf{Enc}}\|_{\text{op}} + \|\dot{\mathbf{Dec}}\|_{\text{op}} \leq \beta \|\dot{x}\|, \quad \beta^2 = \begin{cases} 2dr + 2dK + 2 & \text{for unitaries,} \\ 2dr + 2S + d^2 + 1 & \text{for measurements.} \end{cases} \quad (29)$$

The six coefficients are $1, \sqrt{dr}, \sqrt{dr}, \sqrt{dK}, \sqrt{dK}, 1$ for unitaries and $1, \sqrt{dr}, \sqrt{dr}, \sqrt{S}, \sqrt{S}, d$ for measurements.

Singular values. At $x \in \tilde{\mathcal{Z}}_\delta$, Proposition 4.2 splits the real-linear differential of the overlap map as $d\hat{T}_x = L_x + \Xi_x$, where the local part $L_x(\dot{x}) = -b\hat{T} + \hat{T}a$ takes values in $\mathcal{T}(\hat{T}(x))$, which has dimension at most ℓ by (5), and $\|\Xi_x(\dot{x})\|_F \leq \sqrt{h} \beta \|\dot{x}\| \leq d\delta\beta \|\dot{x}\|$. Directly from $\hat{T} = \text{Dec}^\dagger \text{Enc}$ and $\|Y\|_F \leq d\|Y\|_{\text{op}}$ for maps with domain $\mathbb{C}^{d^2}$, also $\|d\hat{T}_x(\dot{x})\|_F \leq d\beta \|\dot{x}\|$. Dividing by d to pass to the normalized norm, and using Weyl's inequality $\sigma_{\ell+1}(L_x + \Xi_x) \leq \sigma_{\ell+1}(L_x) + \sigma_1(\Xi_x)$ with $\sigma_{\ell+1}(L_x) = 0$, we obtain

$$\sigma_1(d\hat{T}_x) \leq \beta \leq D, \quad \sigma_{\ell+1}(d\hat{T}_x) \leq \delta\beta \leq \delta D. \quad (30)$$

Here $\ell = \ell_u$ or ℓ_m , and $\beta \leq D$ holds for our choice of D .

Ambient extension. So far these are derivatives along the constraint space. The constraint space has empty interior in $\mathbb{R}^D$, while the image-volume lemma below is formulated for C^1 maps on open sets, so we extend $\hat{T}$ to a neighbourhood of $\tilde{\mathcal{Z}}_\delta$ in $\mathbb{R}^D$ without increasing its singular values. Before rescaling, retract a nearby vector by $z \mapsto z/\|z\|$, and a nearby matrix with linearly independent columns by its polar factor $V \mapsto V(V^\dagger V)^{-1/2}$, which has orthonormal columns. At a unit vector z the derivative of the first retraction is $w \mapsto w - z \text{Re}\langle z | w \rangle$, and at an isometry V the derivative of the second is $Y \mapsto Y - \frac{1}{2}V(V^\dagger Y + Y^\dagger V)$. These are the orthogonal projections onto the tangent spaces of the constraints: their outputs satisfy the differentiated constraints, and the subtracted terms are orthogonal to all tangent vectors. Rescaling a block by a scalar preserves this, and dummy coordinates are projected to zero. Composing $\hat{T}$ with the product of these retractions gives a semialgebraic C^1 extension to a neighbourhood of $\tilde{\mathcal{Z}}_\delta$; for instance, the graph of the matrix retraction is locally described by $V = WR$ with $W^\dagger W = I$, $R^\dagger = R$ and $\|R - I\|_F^2 \leq 1/4$, which forces R to be positive definite. At a point of $\tilde{\mathcal{Z}}_\delta$, the extended derivative is $d\hat{T}_x$ composed with an orthogonal projection, so no singular value increases and (30) holds for the ambient derivative. The graph over $\tilde{\mathcal{Z}}_\delta$ is the polynomial graph described above; the extension only supplies the derivative.

We summarize this subsection.

Proposition 5.1 (Witness families). *Let $K \geq 1$ and $0 < \varepsilon < 1$, and let δ be as in (15) and D as in (27). Every target in $\text{Reach}_{K,\varepsilon}$ is within normalized distance δ of $\hat{T}(\tilde{\mathcal{Z}}_\delta)$ for one of at most e^{CD} compact families $\tilde{\mathcal{Z}}_\delta \subseteq \mathbb{R}^D$ of radius at most $\sqrt{6}$. The graph of $\hat{T}$ on each family is described by a fixed number of polynomial equations and inequalities of fixed degrees in $O(D + d^4)$ variables, $\hat{T}$ has a semialgebraic C^1 extension to a neighbourhood of $\tilde{\mathcal{Z}}_\delta$, and on $\tilde{\mathcal{Z}}_\delta$*

$$\sigma_1(d\hat{T}) \leq D, \quad \sigma_{\ell+1}(d\hat{T}) \leq \delta D,$$

with $\ell = \ell_u = 4d^2 - 3$ for unitaries and $\ell = \ell_m = 3d^2 - 2$ for measurements.

5.2 The transverse tube estimate

The geometric input is a quantitative version of the fact that the image of a map whose derivative has rank at most ℓ has dimension at most ℓ . We need it with constants that are exponential in the number of variables and otherwise independent of all dimensions.

Definition 5.2 (Bounded lifted format). *Let $Y \subseteq \mathbb{R}^k$ and let $\phi : Y \rightarrow \mathbb{R}^m$. We say that ϕ has bounded lifted format with j extra variables if its graph is the projection of a compact semialgebraic subset of $\mathbb{R}^{k+m+j}$ that is described by at most Δ_0 polynomial equations and inequalities of degree at most Δ_0 . Here Δ_0 is one absolute constant, fixed once and for all for the finitely many kinds of maps in these notes, and in particular independent of k, m, j and all other dimensions. We write C_Δ for constants that depend only on Δ_0 and on a bound for the radius of Y .*

By Proposition 5.1, every overlap map $\hat{T}$ on a witness family has bounded lifted format with $O(D + d^4)$ variables in total.

Lemma 5.3 (Image volume). *Let $Y \subset \mathbb{R}^k$ be compact and contained in a ball of radius R_0 , and let ϕ be a semialgebraic C^1 map from a neighbourhood of Y to $\mathbb{R}^m$ such that $\phi|_Y$ has bounded lifted format with j extra variables. Then*

$$\text{Vol}_m(\phi(Y)) \leq C_\Delta^{k+m+j} \omega_m \sup_{y \in Y} J_m \phi(y),$$

where ω_m is the volume of the unit ball in $\mathbb{R}^m$ and $J_m \phi(y)$ is the product of the m largest singular values of the derivative of ϕ at y (zero if $k < m$).

The lemma is proved in Appendix A, using the quantitative approximate definable choice of [LRT25] to reduce to an m -dimensional source and the area formula to bound the volume of that source. Applying it to the thickened map $(x, Y') \mapsto \hat{T}(x) + sY'$ gives the following estimate.

Proposition 5.4 (Transverse tube estimate). *Let $\tilde{\mathcal{Z}} \subset \mathbb{R}^D$ be compact and contained in a ball of universal radius, and let $\hat{T} : \tilde{\mathcal{Z}} \rightarrow \mathbb{C}^{d^2 \times d^2}$, with $d \geq 2$ and $N = d^4$, have bounded lifted format with $O(D + N)$ variables and a semialgebraic C^1 extension to a neighbourhood of $\tilde{\mathcal{Z}}$. Suppose that, in normalized Frobenius norm,*

$$\sigma_1(d\hat{T}_x) \leq L, \quad \sigma_{\ell+1}(d\hat{T}_x) \leq \lambda \quad (x \in \tilde{\mathcal{Z}})$$

for some $0 \leq \ell \leq N$. Let $\mathcal{S} \subseteq \text{U}(d^2)$ be measurable with $\text{dist}_n(T, \hat{T}(\tilde{\mathcal{Z}})) \leq \tau$ for all $T \in \mathcal{S}$. If $\tau + \lambda \leq s \leq c_0$ with $s > 0$, then

$$\mu(\mathcal{S}) \leq e^{C_\Delta(D+N)} [C_\Delta(L + s)]^\ell (C_\Delta s)^{N-\ell},$$

where μ is the Haar probability measure and one can take $c_0 = 1/64$. A union of at most e^{CD} such sets only changes the constant.

The mechanism is as follows. Adding a matrix of normalized Frobenius norm at most s to $\hat{T}$ produces an ambient Jacobian with at most ℓ large factors, of size $L + s$, and $2N - \ell$ small factors, of size at most $\lambda + s \leq 2s$. Because $\text{U}(d^2)$ has N normal directions, the s -neighbourhood of $\mathcal{S}$ in the $2N$ -dimensional matrix space has volume at least of order $s^N \mu(\mathcal{S})$, in normalized units; only this lower bound is needed, and Lemma A.1 gives the exact factors. Comparing the two leaves $N - \ell$ small factors. The proof in Appendix A keeps track of the natural scale d of the unitary group in Frobenius norm, so that the threshold c_0 does not depend on d .

For measurements we work with the matrices $T = M^\dagger$. A protocol that witnesses the basis $(\varphi_i)_i$ also witnesses $(e^{i\theta_i} \varphi_i)_i$, after replacing γ_i by $e^{i\theta_i} \gamma_i$, so the sets $\text{Reach}_{K,\varepsilon}^m$ consist of whole fibres $\{\Delta M^\dagger\}$. The diagonal generators $b \in \mathfrak{u}_{\text{diag}}$ in $\mathcal{T}_m$ account for exactly these directions.

5.3 The precision bound for generic targets

Theorem 5.5 (Haar volume at fixed footprint). *There is a universal constant C such that for every $d \geq 2$, $K \geq d^2/4$ and $0 < \varepsilon \leq 1/2$,*

$$\mu(\text{Reach}_{K,\varepsilon}^u) \leq \min\{1, e^{Cd^2K^2} \varepsilon^{\kappa_u/2}\}, \quad \mu(\text{Reach}_{K,\varepsilon}^m) \leq \min\{1, e^{Cd^2K^2} \varepsilon^{\kappa_m/2}\}, \quad (31)$$

with the codimensions $\kappa_u = (d^2 - 1)(d^2 - 3)$ and $\kappa_m = (d^2 - 1)(d^2 - 2)$ of (6). For measurements, $\mu(\text{Reach}_{K,\varepsilon}^m)$ is the invariant measure of the set of approximable PVMs.

Proof. Since $K \geq d^2/4$ we can use $D = \lceil C_* d^2 K^2 \rceil$ for both target classes. Put $\tau = 2\sqrt{\varepsilon}$, which is at least δ in both cases, so every target in $\text{Reach}_{K,\varepsilon}$ is within normalized distance τ of $\hat{T}(\tilde{\mathcal{Z}}_\tau)$ for one of e^{CD} families. Apply Proposition 5.4 to the set $\mathcal{S}$ of targets within distance τ of $\hat{T}(\tilde{\mathcal{Z}}_\tau)$, with the bounds of Proposition 5.1:

$$L = D, \quad \lambda = \tau D, \quad s = \tau + \lambda \leq 2D\tau = 4D\sqrt{\varepsilon}.$$

If $s \leq c_0$, the proposition gives

$$\mu(\text{Reach}_{K,\varepsilon}) \leq e^{C(D+N)}(CD)^\ell (CD\sqrt{\varepsilon})^{N-\ell} \leq e^{C(D+N)}(CD)^N \varepsilon^{\kappa/2}, \quad (32)$$

with ℓ and $\kappa = N - \ell$ for the relevant class. Since $K \geq d^2/4$ and $C_* \geq 16$, we have $D \geq d^6$, hence $N = d^4 \leq D^{2/3}$ and $N \ln(CD) \leq C'D$. All factors other than $\varepsilon^{\kappa/2}$ are therefore at most $e^{CD} \leq e^{C'd^2 K^2}$, which proves (31) in this case.

If $s > c_0$, then $\varepsilon > c_0^2/(16D^2)$, so $\ln(1/\varepsilon) \leq C(1 + \ln D)$ and $\frac{\kappa}{2} \ln(1/\varepsilon) \leq CN(1 + \ln D) \leq C'D$. After enlarging C , the right-hand side of (31) is then at least one. For measurements, the reachable set consists of whole diagonal fibres, so its Haar measure is the invariant measure of its image in $U(d^2)/U(1)^{d^2}$. $\square$

The factor $\varepsilon^{\kappa/2} = (\sqrt{\varepsilon})^\kappa$ agrees with the volume of a tube of radius $\sqrt{\varepsilon}$ around an orbit of codimension κ . The proof does not show that approximable targets are close to exactly implementable ones; it controls the images of the low-leakage witnesses directly. The prefactor $e^{Cd^2 K^2}$ accounts for the $D = O(d^2 K^2)$ parameters.

Corollary 5.6 (Universal approximation). *There are universal constants $c, C > 0$ such that, for $d \geq 2$ and $0 < \varepsilon \leq 1/2$, a footprint K that suffices for every unitary with $\delta_{\text{Choi}} \leq \varepsilon$, or for every rank-one PVM with $\delta_\Phi \leq \varepsilon$, satisfies*

$$K \geq c d \sqrt{\ln(1/\varepsilon)}, \quad \text{equivalently} \quad \varepsilon \geq \exp(-CK^2/d^2).$$

For $d = 2^n$ this reads $\log_2 K \geq n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$.

Proof. The Schmidt-rank floors of Lemma B.1 give $K \geq d^2(1 - \varepsilon) \geq d^2/2$ for all unitaries and $K \geq d^2(1 - \varepsilon)^2 \geq d^2/4$ for all PVMs, so Theorem 5.5 applies. Universality makes the reachable set the whole group, of measure one. Taking logarithms in (31) gives $\frac{\kappa}{2} \ln(1/\varepsilon) \leq Cd^2 K^2$, and $\kappa \geq \frac{3}{16}d^4$ by (6). $\square$

Corollary 5.7 (Almost every fixed target). *There is a universal $c > 0$ such that, for each $d \geq 2$, Haar-almost every $U \in U(d^2)$ has a threshold $\varepsilon_0(U) > 0$ with*

$$K_\varepsilon(U) \geq c d \sqrt{\ln(1/\varepsilon)} \quad (0 < \varepsilon < \varepsilon_0(U)).$$

The same holds for almost every rank-one PVM, with respect to the invariant measure on $U(d^2)/U(1)^{d^2}$.

Proof. Let $\varepsilon_j = e^{-j}$ and $K_j = \lfloor c_1 d \sqrt{j} \rfloor$ with $c_1 > 0$ small. For all large j we have $K_j \geq d^2$, and Theorem 5.5 gives

$$\mu(\text{Reach}_{K_j, \varepsilon_j}) \leq \exp(Cc_1^2 d^4 j - \frac{3}{32} d^4 j) \leq e^{-c_2 d^4 j},$$

which is summable in j . By the Borel–Cantelli lemma, almost every target lies in only finitely many of the sets $\text{Reach}_{K_j, \varepsilon_j}$, so $K_{\varepsilon_j}(U) > K_j$ for all $j \geq j_0(U)$. For $\varepsilon_{j+1} \leq \varepsilon < \varepsilon_j$, monotonicity gives $K_\varepsilon(U) \geq K_{\varepsilon_j}(U) > K_j \geq c d \sqrt{\ln(1/\varepsilon)}$ after decreasing c . $\square$

Remark 5.8 (Parameters against codimension). The witness families have $D = O(d^2 K^2)$ parameters, while the target directions transverse to local equivalence number $\kappa = \Theta(d^4)$. The tube estimate compares $\ln(1/\varepsilon)$ with $D/\kappa = O(K^2/d^2)$, which is the origin of the factor d in Corollaries 5.6 and 5.7. The quantum calculation and this comparison are the same for unitaries and measurements. In the next section we reduce D for targets near SWAP and near a basis of maximally entangled vectors.

6 Universal approximation: targets near SWAP and near a maximally entangled basis

The bound of Corollary 5.6 uses the same witness construction for every target. For universal approximation we can do better by testing the protocol on a fixed ball of targets: for unitaries a ball around SWAP, the unitary that sends Alice's input to Bob's output and vice versa, and for measurements a ball around a basis of maximally entangled vectors. Such targets use both communication directions fully: at small error both messages must have dimension of order d , which leaves Schmidt rank $O(K/d^2)$ for the resource, and the flat Schmidt spectra of these targets let the environment be truncated to total Schmidt rank $O(K/d^2)$ near SWAP and $O(K/d)$ near the maximally entangled basis. This reduces the number of parameters from $O(d^2 K^2)$ to $O(K^2)$. Since we only need to control a constant fraction of the target directions, a pointwise cutoff on the singular values of the leakage residuals suffices for the derivative estimate.

Throughout this section $N = d^4$. In the normalized Frobenius norm, $U(d^2)$ lies in the unit sphere of $\mathbb{R}^{2N}$. Hence, for every $W \in U(d^2)$ and $0 < t \leq 1$,

$$\mu\{V \in U(d^2) : \|V - W\|_n \leq t\} \geq (1 + 2/t)^{-2N}. \quad (33)$$

Indeed, a maximal t -separated subset of $U(d^2)$ has at most $(1 + 2/t)^{2N}$ points, because the balls of radius $t/2$ around its points are disjoint and lie in the ball of radius $1 + t/2$; the balls of radius t around these points cover $U(d^2)$, and by invariance of μ and of the norm under left multiplication they all have the same Haar measure as the ball around W .

6.1 Unitaries

Let

$$\mathcal{B}_d = \{U \in U(d^2) : \|U - \text{SWAP}\|_n \leq 1/4\}.$$

By (33) with $t = \frac{1}{4}$,

$$\mu(\mathcal{B}_d) \geq 9^{-2N}. \quad (34)$$

Lemma 6.1 (Compression near SWAP). *Let $U \in \mathcal{B}_d$ have a protocol of footprint at most K with $\delta_{\text{Choi}} \leq \varepsilon \leq 1/16$, and select a pure resource component with score at least $1 - \varepsilon$, of Schmidt rank r . Then*

$$m_A, m_B \geq d/\sqrt{2}, \quad K \geq d^2/2, \quad r \leq 2K/d^2.$$

Moreover, there is a unit vector $\gamma \in G_A G_B$ with

$$\text{SR}(\gamma) \leq k_\gamma := \lceil K/\lceil d^2/2 \rceil \rceil \leq 4K/d^2, \quad \|F - E_\gamma U\|_n \leq 4\sqrt{\varepsilon}. \quad (35)$$

With these bounds, the witness construction of Section 4.1 uses $D = O(K^2)$ real coordinates.

Proof. Messages. Consider the protocol on the Choi input, and let ρ be the state of Alice's reference $\bar{A}$ together with Bob's kept register K_B after the encoders. Bob's encoder does not touch A or $\bar{A}$, so $\rho = I/d \otimes \sigma$ for a state σ on K_B . For every state on a product $\mathcal{H}_1 \otimes \mathcal{H}_2$ we have $\rho_{12} \leq (\dim \mathcal{H}_2) \rho_1 \otimes I_2$: for a pure state $\sum_j |\psi_j\rangle |j\rangle$ this is Cauchy–Schwarz in the form $|\sum_j \langle \phi_j | \psi_j \rangle|^2 \leq (\dim \mathcal{H}_2) \sum_j |\langle \phi_j | \psi_j \rangle|^2$, and mixed states follow by convexity. Applying this to Bob's incoming message $M_{B_{\text{post}}}$, of dimension m_A , and then Bob's decoding channel gives, for the state of $\bar{A}B'$,

$$\rho_{\bar{A}B'} \leq m_A^2 (I/d \otimes \omega)$$

for some state ω on B' ; the second factor m_A comes from writing $I_{M_{B_{\text{post}}}} = m_A \cdot I_{M_{B_{\text{post}}}}/m_A$. Hence the fidelity of $\rho_{\bar{A}B'}$ with the maximally entangled state of $\bar{A}$ and B' is at most m_A^2/d^2 .

For SWAP this fidelity equals one. The Choi vector of U is within distance $\frac{1}{4}$ of that of SWAP, and the Choi state of the protocol is within trace distance $\sqrt{\varepsilon} \leq \frac{1}{4}$ of that of U , by the Fuchs–van de Graaf inequality. Hence $m_A^2/d^2 \geq \frac{1}{2}$. The same argument applies to m_B , and then $K \geq r m_A m_B \geq r d^2/2$.

Environment. Let $\varpi_1 \geq \varpi_2 \geq \dots$ be the squared Schmidt coefficients of $|U\rangle\rangle$ across the laboratory cut. The Choi vector of SWAP is a product of two maximally entangled states, of $\bar{A}$ with B' and of $\bar{B}$ with A' , so it has d^2 Schmidt coefficients equal to $1/d$. Mirsky's inequality for singular values [Bha97] gives $\sum_i (\sqrt{\varpi_i} - d^{-1})^2 \leq \|U - \text{SWAP}\|_n^2 \leq \frac{1}{16}$. Each index with $\varpi_i < 1/(4d^2)$ contributes more than $1/(4d^2)$ to this sum, so at least $3d^2/4$ indices have $\varpi_i \geq 1/(4d^2)$. Select $k_0 = \lceil d^2/2 \rceil$ of them and put $\alpha_0 = k_0/(4d^2) \geq \frac{1}{8}$. Then $\varpi \geq \alpha_0 \bar{u}$ entrywise, where $\bar{u}$ is the uniform distribution on the selected indices. For every probability vector λ , comparing the mass omitted by the best K products gives

$$1 - \Lambda_K(\varpi \otimes \lambda) \geq \alpha_0 [1 - \Lambda_K(\bar{u} \otimes \lambda)] \geq \alpha_0 [1 - \Lambda_{\lceil K/k_0 \rceil}(\lambda)], \quad (36)$$

where Λ_k denotes the sum of the k largest weights; the second inequality holds because each λ_j/k_0 occurs k_0 times in $\bar{u} \otimes \lambda$.

Now let γ' and the score $p = q_U \geq 1 - \varepsilon$ be as in the proof of Lemma 3.2(i), and let λ be the Schmidt spectrum of γ' . The Choi vector of F has Schmidt rank at most K and overlap $\sqrt{p}$ with $|U\rangle\rangle \otimes \gamma'$, whose spectrum is $\varpi \otimes \lambda$, so $\Lambda_K(\varpi \otimes \lambda) \geq p$. By (36), the top- k_γ truncation of γ' retains mass $t \geq 1 - 8\varepsilon$. Let γ be this truncation, normalized. As in Lemma 3.2, the overlap of $|F\rangle\rangle$ with $|U\rangle\rangle \otimes \gamma$ is $\sqrt{pt} \geq \min\{p, t\} \geq 1 - 8\varepsilon$, so $\|F - E_\gamma U\|_n^2 \leq 16\varepsilon$. Finally $k_\gamma \leq K/\lceil d^2/2 \rceil + 1 \leq 2K/d^2 + 1 \leq 4K/d^2$, since $K \geq d^2/2$.

Coordinates. Since $r m_X \leq K/m_{X^c} \leq \sqrt{2}K/d$, where X^c is the other party, each encoder uses $O(d^2 r^2 m_X^2) = O(K^2)$ coordinates. The support spaces are $\hat{G}_X = X' \otimes \mathbb{C}^{k_\gamma}$, of dimension dk_γ , and Q_X has dimension at most dK , so each completed isometry J_X uses $O(d^2 K k_\gamma + d^2 k_\gamma^2) = O(K^2)$ coordinates. The resource and γ use $O(r^2 + k_\gamma^2) = O(K^2)$. $\square$

Lemma 6.2 (Most target directions move slowly). *The witnesses of Lemma 6.1, with $\delta = 4\sqrt{\varepsilon}$, admit coordinates of universal radius and ambient extensions as in Section 5.1 for which, in normalized Frobenius norm,*

$$\sigma_1(d\hat{T}) \leq C\sqrt{K/d^2}, \quad \sigma_{\ell_\star+1}(d\hat{T}) \leq C\delta\sqrt{K/d^2}, \quad \ell_\star := 4d^2 - 3 + 4d^2 \lfloor d^2/64 \rfloor \leq \frac{7}{8}N. \quad (37)$$

Proof. Divide each isometry block with k columns by $\sqrt{k}$ and leave the vectors unchanged. We now bound Frobenius norms of the velocities of Enc and Dec, instead of operator norms. For the encoders, the compression lemma of [FGMS26] gives

$$\|(\dot{V}_A \otimes V_B)E_\eta\|_F^2 = d \operatorname{Tr}[\dot{V}_A^\dagger \dot{V}_A (I_d \otimes \eta_A)] \leq d \|\dot{V}_A\|_F^2,$$

where η_A is the reduced state of the resource. In the scaled coordinates $\dot{V}_A$ is $\sqrt{dr}$ times the coordinate velocity, so this term contributes at most $d\sqrt{r}$ times the coordinate speed. In the same way the completed isometries contribute at most $d\sqrt{k_\gamma}$, and the two vectors at most d . Since $r, k_\gamma \leq 4K/d^2$ and $d^2 \leq 2K$,

$$\|d\text{Enc}(v)\|_F + \|d\text{Dec}(v)\|_F \leq C\sqrt{K} \|v\|. \quad (38)$$

The same bound holds for $d\hat{T}(v) = d\text{Dec}(v)^\dagger \text{Enc} + \text{Dec}^\dagger d\text{Enc}(v)$, which gives the first bound in (37) after dividing by d .

Both leakage residuals have $\|Y_{\text{Enc}}\|_F^2 = \|Y_{\text{Dec}}\|_F^2 = h \leq d^2\delta^2$. Split each of them at singular value 8δ into a low part, of operator norm at most 8δ , and a high part, of rank at most $\lfloor d^2/64 \rfloor$. In the residual velocity $\Xi = d\text{Dec}(v)^\dagger Y_{\text{Enc}} + Y_{\text{Dec}}^\dagger d\text{Enc}(v)$, the high part of Y_{Enc} has

a right support of dimension at most $\lfloor d^2/64 \rfloor$ that does not depend on v , and the high part of $Y_{\text{Dec}}^\dagger$ has a left support of the same dimension. As real-linear maps of v , the two high contributions therefore take values in a fixed space of real dimension at most $4d^2 \lfloor d^2/64 \rfloor$. Together with the allowed directions, of dimension at most $4d^2 - 3$, this gives a space of dimension at most $\ell_\star$. Let Π be the orthogonal projection onto this space. Then $\Pi d\hat{T}$ has rank at most $\ell_\star$, and $(I - \Pi) d\hat{T}(v)$ is the projection of the low contributions, of Frobenius norm at most $8\delta(\|d\text{Dec}(v)\|_F + \|d\text{Enc}(v)\|_F) \leq C\delta\sqrt{K}\|v\|$ by (38). Dividing by d and using $\sigma_{\ell_\star+1}(d\hat{T}) \leq \|d\hat{T} - \Pi d\hat{T}\|_{\text{op}}$, which holds because $\Pi d\hat{T}$ has rank at most $\ell_\star$, proves the second bound. The split is pointwise bookkeeping and need not vary smoothly. The ambient extension of Section 5.1 precomposes the derivative with an orthogonal projection, so the bounds survive. Finally, $(4d^2 - 3)/d^4 \leq 13/16$ and $4d^2 \lfloor d^2/64 \rfloor \leq N/16$ for $d \geq 2$. $\square$

Theorem 6.3 (Haar volume near SWAP). *There is a universal C such that for $d \geq 2$, $K \geq d^2/2$ and $0 < \varepsilon \leq 1/2$,*

$$\mu(\mathcal{B}_d \cap \text{Reach}_{K,\varepsilon}^u) \leq \min\{1, e^{CK^2} \varepsilon^{d^4/16}\}.$$

Proof. First let $\varepsilon \leq 1/16$, so $\delta = 4\sqrt{\varepsilon} \leq 1$. The witness families of Lemma 6.1 have $D = O(K^2)$ coordinates, bounded lifted format, and at most e^{CK^2} members, and every target is within normalized distance δ of an image. Put $u = K/d^2 \geq \frac{1}{2}$ and apply Proposition 5.4 with the bounds of Lemma 6.2, with $\ell = \ell_\star$, $\tau = \delta$, and $s = C_2\delta\sqrt{1+u}$ for a constant C_2 large enough that $s \geq \tau + \lambda$. If $s \leq c_0$, then

$$\mu(\mathcal{B}_d \cap \text{Reach}_{K,\varepsilon}^u) \leq e^{C(K^2+N)} [C\sqrt{1+u}]^N \delta^{N-\ell_\star} \leq e^{C'K^2} \varepsilon^{N/16},$$

because $N - \ell_\star \geq N/8$, $\delta^{N/8} = (16\varepsilon)^{N/16}$, $N \leq 4K^2$, and $N \ln(1+u) \leq Nu \leq 2K^2$. If $s > c_0$, then $\ln(1/\varepsilon) \leq C + \ln(1+u)$, and the claimed bound is at least one after enlarging C . The same holds for $1/16 < \varepsilon \leq 1/2$. $\square$

The exponent $N/16$ is smaller than the exponent $\kappa_u/2 \approx N/2$ of Theorem 5.5; the gain is in the prefactor, e^{CK^2} in place of $e^{Cd^2K^2}$, and it comes from the smaller witness families.

Corollary 6.4 (Universal unitary approximation). *There are universal $c, C > 0$ such that, for $d \geq 2$ and $0 < \varepsilon \leq 1/2$, a footprint K that suffices for every unitary with $\delta_{\text{Choi}} \leq \varepsilon$ satisfies*

$$K \geq c d^2 \sqrt{\ln(1/\varepsilon)}, \quad \text{equivalently} \quad \varepsilon \geq \exp(-CK^2/d^4), \quad \text{and} \quad K \geq d^2(1 - \varepsilon).$$

For $d = 2^n$, $\log_2 K \geq 2n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$.

Proof. The Schmidt-rank floor for SWAP (Lemma B.1) gives $K \geq d^2(1 - \varepsilon) \geq d^2/2$. Universality puts $\mathcal{B}_d$ inside the reachable set, so (34) and Theorem 6.3 give $9^{-2N} \leq e^{CK^2} \varepsilon^{N/16}$. Taking logarithms, $\frac{N}{16} \ln(1/\varepsilon) \leq CK^2 + 2N \ln 9 \leq C'K^2$, and $N = d^4$. $\square$

6.2 Measurements

For measurements we replace SWAP by a basis of maximally entangled vectors. Let $|\Phi^+\rangle = d^{-1/2} \sum_a |a\rangle|a\rangle$, let $W_1, \dots, W_{d^2}$ be the Weyl operators (generalized Pauli operators) on $\mathbb{C}^d$, and let $M_{\text{Bell}} = \sum_i |e_i\rangle\langle i|$ be the basis matrix of the maximally entangled basis $e_i = (I \otimes W_i)|\Phi^+\rangle$. Put

$$t_d = \frac{1}{4d^{3/2}}, \quad \mathcal{M}_d = \{M \in \text{U}(d^2) : \|M - M_{\text{Bell}}\|_{\text{n}} \leq t_d\}, \quad \mathcal{M}_d^\dagger = \{M^\dagger : M \in \mathcal{M}_d\}.$$

The ball is much smaller than $\mathcal{B}_d$, but by (33) and the inversion invariance of μ ,

$$\mu(\mathcal{M}_d^\dagger) = \mu(\mathcal{M}_d) \geq (1 + 8d^{3/2})^{-2N}, \quad (39)$$

which is still $e^{-O(N \ln d)}$. Its radius is chosen so that every basis vector in $\mathcal{M}_d$ has an almost flat Schmidt spectrum.

Lemma 6.5 (Compression near a maximally entangled basis). *Let $M \in \mathcal{M}_d$ have a protocol of footprint at most K with $\delta_\Phi \leq \varepsilon \leq 1/256$, and select a pure resource component with score at least $1 - \varepsilon$, of Schmidt rank r . Then*

$$m_A, m_B \geq d/\sqrt{2}, \quad K \geq d^3/2, \quad r \leq 2K/d^2.$$

Moreover, there are unit vectors $\gamma_1, \dots, \gamma_{d^2} \in G_A G_B$ with

$$\sum_i \text{SR}(\gamma_i) \leq \lceil K/d \rceil + d^2 \leq 4K/d, \quad \|F - C_\gamma M^\dagger\|_n \leq \sqrt{6\varepsilon}. \quad (40)$$

With these bounds, the witness construction of Section 4.1 uses $D = O(K^2)$ real coordinates.

Proof. Flat spectra. Let φ_i be the columns of M . Then $\|\varphi_i - e_i\| \leq \|M - M_{\text{Bell}}\|_F \leq dt_d = \frac{1}{4}d^{-1/2}$. Reshaping a vector of $\mathbb{C}^d \otimes \mathbb{C}^d$ into its $d \times d$ coefficient matrix preserves the Euclidean norm, the coefficient matrix of e_i has all singular values equal to $d^{-1/2}$, and singular values are 1-Lipschitz in the Frobenius norm [Bha97]. Hence all d Schmidt coefficients of φ_i lie in $[\frac{3}{4}d^{-1/2}, \frac{5}{4}d^{-1/2}]$, and the squared ones satisfy

$$\frac{9}{16d} \leq \lambda_{ij} \leq \frac{25}{16d} \quad (1 \leq j \leq d). \quad (41)$$

Footprint. We use the notation of the proof of Lemma 3.2(ii), with $p = q_\Phi \geq 1 - \varepsilon$. Since $\zeta_{ik}^2 \leq \|\tilde{\gamma}_i\|^2 \leq 1$, every squared Schmidt coefficient (17) of v_0 is at most $25/(16d^3)$, so $p^2 \leq \Lambda_K(v_0) \leq 25K/(16d^3)$ and $K \geq \frac{16}{25}(1 - \varepsilon)^2 d^3 \geq d^3/2$.

Messages. Run the protocol on the Choi input, in which A is maximally entangled with Alice's reference $\bar{A}$ and B with Bob's reference $\bar{B}$. For the frozen dilation $C_\gamma M^\dagger$ of Lemma 3.2(ii), both parties output the same label i , with probability d^{-2} , and the references are then left in the state $|\bar{\varphi}_i\rangle$, since $|\Omega\rangle = d^{-1} \sum_i |\varphi_i\rangle |\bar{\varphi}_i\rangle$. Let Bob, who holds $\bar{B}$ and his label i , apply $\bar{W}_i^\dagger$ to $\bar{B}$. Since $|\bar{e}_i\rangle = (I \otimes \bar{W}_i) |\Phi^+\rangle$, the pair $\bar{A}\bar{B}$ then has fidelity $|\langle e_i | \varphi_i \rangle|^2 \geq 1 - \|\varphi_i - e_i\|^2$ with $|\Phi^+\rangle$, and on average over i at least $1 - t_d^2$. For the actual protocol F , whose Choi vector is within $\|F - C_\gamma M^\dagger\|_n \leq 2\sqrt{\varepsilon}$ of that of $C_\gamma M^\dagger$, the same local correction by Bob gives fidelity at least $1 - t_d^2 - 2\sqrt{\varepsilon} \geq \frac{1}{2}$, because trace distance between pure states is at most their Euclidean distance and channels do not increase it. On the other hand, before Bob receives Alice's message, the state of $\bar{A}$ together with Bob's kept register and $\bar{B}$ is $I/d \otimes \sigma$, because $\bar{A}$ is correlated only with Alice's input and Alice's operations do not change this reduced state. Exactly as in the proof of Lemma 6.1, with $\bar{B}$ among Bob's registers, the final state of $\bar{A}\bar{B}$ is bounded by $m_A^2(I/d \otimes \omega)$ for a state ω , so its fidelity with $|\Phi^+\rangle$ is at most m_A^2/d^2 . Hence $m_A^2 \geq d^2/2$. The same argument with the roles of Alice and Bob exchanged, Alice correcting $\bar{A}$ with her label, gives $m_B^2 \geq d^2/2$, and then $r \leq K/(m_A m_B) \leq 2K/d^2$.

Environment. Let $a_{ik} = d^{-2} \zeta_{ik}^2$, so that $\sum_{i,k} a_{ik} = p$ and the squared Schmidt coefficients of v_0 are $w_{ijk} = a_{ik} \lambda_{ij}$. For a nonnegative vector x let $\Lambda_k(x)$ be the sum of its k largest entries. If a set $\mathcal{E}$ of K triples (i, j, k) carries the top- K mass of v_0 , then, by (41),

$$p - \Lambda_K(v_0) = \sum_{(i,j,k) \notin \mathcal{E}} w_{ijk} \geq \frac{9}{16} \sum_{(i,j,k) \notin \mathcal{E}} \frac{a_{ik}}{d} \geq \frac{9}{16} [p - \Lambda_{\lceil K/d \rceil}(a)],$$

where the last step holds because each pair (i, k) occurs in at most d triples, so the K triples of $\mathcal{E}$ carry at most the mass $\Lambda_{\lceil K/d \rceil}(a)$ of the vector $(a_{ik}/d)_{i,j,k}$. With $\Lambda_K(v_0) \geq p^2$ this gives $p - \Lambda_{\lceil K/d \rceil}(a) \leq \frac{16}{9}(p - p^2) \leq \frac{16}{9}\varepsilon$. Now keep in each $\tilde{\gamma}_i$ only the Schmidt terms k for which a_{ik} is among the $\lceil K/d \rceil$ largest entries of a , and define t_i and γ_i from these terms as in the proof of

Lemma 3.2(ii). Then $\sum_i \text{SR}(\gamma_i) \leq \lceil K/d \rceil + d^2$ and $d^{-2} \sum_i t_i^2 = \Lambda_{\lceil K/d \rceil}(a) \geq p - \frac{16}{9}\varepsilon \geq 1 - 3\varepsilon$, so the overlap estimate of that proof gives $\|F - C_\gamma M^\dagger\|_n^2 \leq 6\varepsilon$. Since $K \geq d^3/2$, we have $d^2 \leq 2K/d$ and $1 \leq K/d$, so $\lceil K/d \rceil + d^2 \leq 4K/d$.

Coordinates. Since $r m_X \leq K/m_{X^c} \leq \sqrt{2}K/d$, where X^c is the other party, each encoder uses $O(d^2 r^2 m_X^2) = O(K^2)$ coordinates, and the resource uses $O(r^2) = O(K^2)$. With $S = \sum_i \text{SR}(\gamma_i) \leq 4K/d$, the vectors γ_i use $O(S^2)$ coordinates and each completed isometry J_X , from $\widehat{G}_X$ of dimension S to $Q_X \oplus \widehat{G}_X$ of dimension at most $dK + S$, uses $O((dK + S)S) = O(K^2)$ coordinates. There are at most K^3 factorizations and at most $\binom{\lceil K/d \rceil + d^2}{d^2} \leq 2^{\lceil K/d \rceil + d^2} \leq 2^{3K}$ rank allocations. $\square$

Lemma 6.6 (Most target directions move slowly). *The witnesses of Lemma 6.5, with $\delta = \sqrt{6\varepsilon}$, admit coordinates of universal radius and ambient extensions as in Section 5.1 for which the bounds (37) hold, with the same $\ell_\star$.*

Proof. Use the scaling of Section 5.1 for measurements. As in the proof of Lemma 6.2, the encoders contribute at most $d\sqrt{r}$ times the coordinate speed to $\|\text{dEnc}(v)\|_F$, and the resource at most d . For the completed isometries,

$$\|(\text{d}J_A \otimes J_B)C_\gamma\|_F^2 = \text{Tr}[\text{d}J_A^\dagger \text{d}J_A \rho_A] \leq \|\text{d}J_A\|_F^2, \quad \rho_A = \sum_i |i\rangle\langle i| \otimes \gamma_{i,A},$$

where $\gamma_{i,A}$ is the reduced state of γ_i on Alice's factor; ρ_A is block diagonal with density matrices as blocks, so $\|\rho_A\|_{\text{op}} \leq 1$; in the scaled coordinates this is at most $\sqrt{S}$ times the coordinate speed, and the same holds for J_B . The tuple $(\gamma_i)_i$ contributes $\|\text{d}C_\gamma\|_F = (\sum_i \|\dot{\gamma}_i\|^2)^{1/2}$, which is d times its coordinate speed. Since $r \leq 2K/d^2$, $S \leq 4K/d$ and $d^2 \leq 2K/d \leq K$, all these coefficients are at most $2\sqrt{K}$, so (38) holds. The rest of the proof of Lemma 6.2 applies verbatim, because the allowed directions for measurements have dimension at most $\ell_m = 3d^2 - 2 \leq 4d^2 - 3$. $\square$

Theorem 6.7 (Haar volume near a maximally entangled basis). *There is a universal C such that for $d \geq 2$, $K \geq d^3/4$ and $0 < \varepsilon \leq 1/2$,*

$$\mu(\mathcal{M}_d^\dagger \cap \text{Reach}_{K,\varepsilon}^m) \leq \min\{1, e^{CK^2} \varepsilon^{d^4/16}\}.$$

Proof. First let $\varepsilon \leq 1/256$, so $\delta = \sqrt{6\varepsilon} \leq 1$. If $K < d^3/2$ the set is empty by Lemma 6.5, so let $K \geq d^3/2$. We run the construction of Section 4.1 with the vectors γ_i of (40) in place of those of Lemma 3.2(ii), and with $\delta = \sqrt{6\varepsilon}$ in (22) and (24). The resulting witness families have $D = O(K^2)$ coordinates, bounded lifted format, and at most e^{CK^2} members, and every target is within normalized distance δ of an image, by (24). Put $u = K/d^2 \geq d/2$ and apply Proposition 5.4 with the bounds of Lemma 6.6, as in the proof of Theorem 6.3. If $s \leq c_0$, this gives

$$\mu(\mathcal{M}_d^\dagger \cap \text{Reach}_{K,\varepsilon}^m) \leq e^{C(K^2+N)} [C\sqrt{1+u}]^N \delta^{N/8} \leq e^{C'K^2} \varepsilon^{N/16},$$

because $N \leq K^2$ and $N \ln(1+u) \leq Nu = d^2 K \leq K^2$. If $s > c_0$, then $\ln(1/\varepsilon) \leq C + \ln(1+u)$, and the claimed bound is at least one after enlarging C . For $1/256 < \varepsilon \leq 1/2$ the bound is at least one once $C \geq 2$, since $K \geq d^3/4$ gives $N \leq 4K^2$. $\square$

Corollary 6.8 (Universal measurement approximation). *There are universal $c, C > 0$ such that, for $d \geq 2$ and $0 < \varepsilon \leq 1/2$, a footprint K that suffices for every rank-one PVM with $\delta_\Phi \leq \varepsilon$ satisfies*

$$K \geq c d^2 \sqrt{\ln(1/\varepsilon)}, \quad \text{equivalently} \quad \varepsilon \geq \exp(-CK^2/d^4), \quad \text{and} \quad K \geq d^3(1-\varepsilon)^2.$$

For $d = 2^n$, $\log_2 K \geq \max\{3n - 2, 2n + \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)\}$.

Proof. The floor for a basis of maximally entangled vectors (Lemma B.1) gives $K \geq d^3(1 - \varepsilon)^2 \geq d^3/4$. Universality puts $\mathcal{M}_d^\dagger$ inside the reachable set, so (39) and Theorem 6.7 give $(1 + 8d^{3/2})^{-2N} \leq e^{CK^2} \varepsilon^{N/16}$. Taking logarithms, $\frac{N}{16} \ln(1/\varepsilon) \leq CK^2 + 2N \ln(1 + 8d^{3/2}) \leq C'K^2$, since $K^2 \geq d^6/16$ absorbs $N \ln d$. $\square$

The measurement bound thus has the same precision term as the unitary bound, and a larger floor. In the last step, the small radius t_d costs a factor $\ln d$ in the logarithm of the mass of the ball, which the floor $K \geq d^3/4$ absorbs.

6.3 Consequences and limits

Both regional estimates also give pointwise bounds for almost every target in their balls. Let $\varepsilon_j = e^{-j}$ and $K_j = \lfloor c d^2 \sqrt{j} \rfloor$ with $c > 0$ small. For all large j , $K_j \geq d^2/2$ and $K_j \geq d^3/4$, so Theorems 6.3 and 6.7 bound the measures of $\mathcal{B}_d \cap \text{Reach}_{K_j, \varepsilon_j}^u$ and $\mathcal{M}_d^\dagger \cap \text{Reach}_{K_j, \varepsilon_j}^m$ by $e^{-(1/16 - Cc^2)d^4 j}$, which is summable. The Borel–Cantelli argument of Corollary 5.7 then shows that, for Haar-almost every $U \in \mathcal{B}_d$ and for the PVM of Haar-almost every basis matrix $M \in \mathcal{M}_d$, there is a threshold $\varepsilon_0 > 0$ below which $K_\varepsilon \geq c d^2 \sqrt{\ln(1/\varepsilon)}$, after decreasing c . The balls have positive but very small measure, so this does not improve Corollary 5.7, whose factor d comes from the whole-group estimate of Theorem 5.5. Neither regional estimate strengthens that whole-group estimate.

7 Free classical communication

The footprint charges classical messages, and this is what gives the strong dimension dependence of Theorems A and B. It is not needed for divergence with the precision. In this section classical communication is free: the classical messages may range over arbitrary, even continuous, alphabets, and we charge only the Schmidt number of the resource and the dimensions of the quantum messages.

Model. Every quantum register, including the quantum messages, has a fixed finite dimension. Party $X \in \{A, B\}$ first applies, to its input and its share of the resource, an instrument with outcomes in a standard Borel space Σ_X : a countably additive map $E \mapsto \mathcal{I}_X(E)$ from Borel sets $E \subseteq \Sigma_X$ to completely positive maps between fixed finite-dimensional matrix algebras, with $\mathcal{I}_X(\Sigma_X)$ trace preserving. Its quantum output consists of a kept register and an outgoing quantum message of dimension m_X^q . Both parties then send their quantum messages and broadcast their classical outcomes $x \in \Sigma_A$ and $y \in \Sigma_B$ simultaneously, and apply final channels that may depend on (x, y) ; we require their Choi matrices to be jointly Borel measurable in (x, y) . We charge the *quantum footprint*

$$K_q = r m_A^q m_B^q.$$

Local operations and simultaneous classical communication (LOSCC [GAVC25], also called LOBC [GC20]) is the case $m_A^q = m_B^q = 1$, where $K_q = r$. Let $\text{Reach}_{K_q, \varepsilon}^{q,u}$ and $\text{Reach}_{K_q, \varepsilon}^{q,m}$ be the unitaries, respectively the matrices $M^\dagger$ of PVMs, that such protocols of quantum footprint at most K_q implement with $\delta_{\text{Choi}} \leq \varepsilon$, respectively $\delta_\Phi \leq \varepsilon$; as before we drop the superscripts u and m when a statement applies to both. We write $K_{q, \varepsilon}(T)$ for the least quantum footprint of a target T at error ε . We use the Haar outer measure μ^* , since this class of protocols has no finite-dimensional parametrization. By Lemma 7.1 below, each of these reachable sets is contained in a compact reachable set $\text{Reach}_{\overline{K}, \varepsilon}$ of the footprint model, and the measure bounds below only use this containment.

Lemma 7.1 (Finite-support compression of the score). *Consider a protocol in this model whose resource has Schmidt number at most r , and fix a target unitary or PVM with its score q_U or q_Φ . There is a protocol with*

- (i) *a pure resource of Schmidt rank $r' \leq r$;*
- (ii) *the same quantum-message dimensions m_A^q, m_B^q ;*
- (iii) *finite classical alphabets $\Sigma'_X \subseteq \Sigma_X$ of sizes $|\Sigma'_X| \leq (dr')^2 + 1$;*
- (iv) *a score at least that of the original protocol.*

Copying the finite classical outcomes coherently into the messages turns it into a protocol of footprint at most

$$\overline{K} := 4d^4 K_q^5. \quad (42)$$

Proof. The score is affine in the resource, so we select a pure component whose score is at least the average; let r' be its Schmidt rank and restrict the resource registers to the Schmidt supports. The input of party X 's instrument then has dimension $s_X = dr'$.

We first compress Alice's outcomes. Let $J_A(E) = \sum_{j,k} |j\rangle\langle k| \otimes \mathcal{I}_A(E)(|j\rangle\langle k|)$ be the unnormalized Choi matrix of the branch E . Then $\mu_A(E) = \text{Tr } J_A(E)/s_A$ is a probability measure, and finite-dimensional Radon–Nikodym gives a positive density $c_A(x)$ with $J_A(E) = \int_E c_A d\mu_A$ and $\text{Tr } c_A(x) = s_A$ almost everywhere. Let $\bar{c}_A(x)$ be the partial trace of $c_A(x)$ over the output; then $\text{Tr } \bar{c}_A(x) = s_A$ and $\int \bar{c}_A d\mu_A = I$, which expresses trace preservation. Hold Bob's instrument and all final channels fixed, and let $f_A(x)$ be the contribution of Alice's branch x to the score, after averaging over Bob's outcome. It is measurable by Fubini's theorem, and $0 \leq f_A(x) \leq \|\bar{c}_A(x)\|_{\text{op}} \leq s_A$, since a contribution to the score is at most the probability of its branch. The pairs $(\bar{c}_A(x), f_A(x))$ lie in a real affine space of dimension $(s_A^2 - 1) + 1 = s_A^2$, and their μ_A -average is (I, q) , where q is the score. By Carathéodory's theorem there are outcomes $x_1, \dots, x_{N_A}$ with $N_A \leq s_A^2 + 1$ and probabilities p_j such that

$$\sum_j p_j \bar{c}_A(x_j) = I, \quad \sum_j p_j f_A(x_j) = q.$$

The first identity says that the branches $p_j c_A(x_j)$, with the final channels of the outcomes x_j , form a trace-preserving instrument; the second says that the score is unchanged.

The selected values have to be attained at actual outcomes. Fix by Fubini a set Σ_A^0 of full μ_A -measure on which the density is defined and the required identities for Bob's outcomes hold, and make all selections in Σ_A^0 . *Claim: the average (I, q) is a convex combination of at most $s_A^2 + 1$ values $(\bar{c}_A(x), f_A(x))$ with $x \in \Sigma_A^0$.* The average lies in the closed convex hull of these values. If it lies in the relative interior of that hull, finitely many attained values surround it, and Carathéodory reduces their number to at most $s_A^2 + 1$. If it lies on the relative boundary, a supporting hyperplane through it contains almost all values, and the claim follows by induction on the affine dimension.

With Alice's finite instrument fixed, we perform the same selection for Bob inside the intersection of the finitely many full-measure sets associated with $x_1, \dots, x_{N_A}$. This gives (iii) for Bob and preserves the score exactly; in the pure selected component the score does not decrease, which gives (iv). This ordered choice is where joint measurability is used.

Each selected branch is a completely positive map with a finite Kraus representation, so its label can be copied coherently into one kept and one sent register. The complete outgoing messages then have dimensions $m_X^q |\Sigma'_X|$. With $|\Sigma'_X| \leq 2d^2 r'^2$ and $r' \leq r' m_A^q m_B^q \leq K_q$, the footprint is

$$r'(m_A^q |\Sigma'_A|)(m_B^q |\Sigma'_B|) \leq 4d^4 r'^4 \cdot r' m_A^q m_B^q \leq 4d^4 K_q^5. \quad \square$$

The compression depends on the target: it preserves the score, not the whole channel. This suffices because every lower bound in these notes only uses the score (Remark 3.3).

The Schmidt-rank floors of Lemma B.1 survive free classical communication in a weaker form, because classical messages carry no Schmidt rank across the cut once they are conditioned on.

Lemma 7.2 (Floors with free classical communication). *Consider a protocol with free classical messages and quantum footprint at most K_q .*

- (i) *For a unitary target U , $q_U \leq \Lambda_{K_q}(|U\rangle\rangle)$.*
- (ii) *For a PVM with basis $(\varphi_i)_{i \leq d^2}$, $q_\Phi \leq \max_i \Lambda_{K_q}(\varphi_i)$.*

Consequently, approximating SWAP with $\delta_{\text{Choi}} \leq \varepsilon$ requires $K_q \geq d^2(1 - \varepsilon)$, and approximating the measurement in a basis of maximally entangled vectors with $\delta_\Phi \leq \varepsilon$ requires $K_q \geq d(1 - \varepsilon)$. Almost every unitary U requires $K_q \geq d^2$ when $\varepsilon < \varpi_{\min}(U)$, and almost every PVM requires $K_q \geq d$ when $\varepsilon < \lambda_*$, with $\varpi_{\min}$ and λ_* as in Lemma B.1(iii).

Proof. By Lemma 7.1, which does not decrease the score and keeps the quantum-message dimensions, we may assume a pure resource of Schmidt rank at most r and finite classical alphabets. Run the protocol on the Choi input and fix a pair of classical outcomes and Kraus operators of all local operations, including the discarding of garbage. The resulting unnormalized vector on $A'\bar{A}$ and $B'\bar{B}$ has Schmidt rank at most $r m_A^q m_B^q \leq K_q$ across the laboratory cut: as in Lemma 3.1, only the quantum messages cross the cut, and the classical outcomes are fixed. The Choi state ρ of the implemented channel is the finite sum of the rank-one operators of these vectors, so it has Schmidt number at most K_q . (i) Hence $q_U = \langle\langle U|\rho|U\rangle\rangle \leq \Lambda_{K_q}(|U\rangle\rangle)$ by (11). (ii) Let $\sigma_i = (\langle i|_{A'} \langle i|_{B'} \otimes I) \rho (|i\rangle_{A'} |i\rangle_{B'} \otimes I)$ on $\bar{A}\bar{B}$. Projecting onto a product vector is a local operation, so each σ_i has Schmidt number at most K_q , and $\sum_i \text{Tr } \sigma_i \leq 1$. Since $|\Omega\rangle = d^{-1} \sum_i |\varphi_i\rangle |\bar{\varphi}_i\rangle$,

$$q_\Phi = \sum_i \langle \bar{\varphi}_i | \sigma_i | \bar{\varphi}_i \rangle \leq \sum_i \text{Tr}(\sigma_i) \Lambda_{K_q}(\varphi_i) \leq \max_i \Lambda_{K_q}(\varphi_i).$$

The consequences follow as in Lemma B.1: $\Lambda_k(|\text{SWAP}\rangle\rangle) = \min\{k/d^2, 1\}$, a maximally entangled vector has $\Lambda_k = \min\{k/d, 1\}$, and if K_q is below the Schmidt rank then Λ_{K_q} is at most one minus the smallest squared Schmidt coefficient. $\square$

With classical messages free, the labels no longer need to be charged, which is why the floor for measurements drops from d^3 to d .

Corollary 7.3 (Free classical communication). *There are universal $c, C > 0$ such that, for $d \geq 2$, $K_q \geq 1$ and $0 < \varepsilon \leq 1/2$,*

$$\mu^*(\text{Reach}_{K_q, \varepsilon}^{q, u}) \leq \min\{1, e^{C d^{10} K_q^{10} \varepsilon^{\kappa_u/2}}\}, \quad \mu^*(\text{Reach}_{K_q, \varepsilon}^{q, m}) \leq \min\{1, e^{C d^{10} K_q^{10} \varepsilon^{\kappa_m/2}}\}. \quad (43)$$

Consequently:

- (i) *a quantum footprint that suffices for every unitary, respectively for every rank-one PVM, satisfies*

$$\begin{aligned} K_q &\geq \max\{d^2(1 - \varepsilon), c d^{-2/5} [\ln(1/\varepsilon)]^{1/10}\} \quad (\text{unitaries}), \\ K_q &\geq \max\{d(1 - \varepsilon), c d^{-2/5} [\ln(1/\varepsilon)]^{1/10}\} \quad (\text{PVMs}); \end{aligned}$$

for $d = 2^n$, in both cases $\log_2 K_q \geq \frac{1}{10} \log_2 \ln(1/\varepsilon) - \frac{2}{5}n - O(1)$;

- (ii) *for each $d \geq 2$, almost every unitary and almost every rank-one PVM T has a threshold $\varepsilon_0(T) > 0$ such that $K_{q, \varepsilon}(T) \geq c d^{-3/5} [\ln(1/\varepsilon)]^{1/10}$ for $0 < \varepsilon < \varepsilon_0(T)$.*

In LOSCC, $K_q = r$, and a resource of Schmidt number r occupies at least $2 \log_2 r$ qubits.

Proof. By Lemma 7.1, $\text{Reach}_{K_q, \varepsilon}^q$ is contained in the compact set $\text{Reach}_{\bar{K}, \varepsilon}$, where $\bar{K} = 4d^4 K_q^5 \geq d^2$. Theorem 5.5 with $d^2 \bar{K}^2 \leq 16d^{10} K_q^{10}$ gives (43). For (i), the floors are those of Lemma 7.2; for the precision terms, universality in this model gives universality in the footprint model at footprint $\bar{K}$, and Corollaries 6.4 and 6.8 give $c d^2 \sqrt{\ln(1/\varepsilon)} \leq \bar{K} = 4d^4 K_q^5$. For (ii), let $\varepsilon_j = e^{-j}$ and $K_j = \lfloor c_1 d^{-3/5} j^{1/10} \rfloor$ with c_1 small; for large j , $K_j \geq 1$ and $d^{10} K_j^{10} \leq c_1^{10} d^4 j$, so the bounds (43) are at most $e^{-c_2 d^4 j}$. Borel–Cantelli for outer measures and monotonicity in K_q and ε conclude as in Corollary 5.7. $\square$

Localizable measurements. In an *approximate localization* of a PVM Φ , Alice and Bob share a finite-dimensional state of Schmidt number r , do not communicate, and measure their inputs together with their shares of the state, with outcomes x and y in standard Borel spaces, for instance finite sets. A fixed measurable function $f(x, y) \in \{1, \dots, d^2\}$ is the reported outcome, and the failure probability is $1 - d^{-2} \sum_i \Pr[f(x, y) = i \mid \varphi_i]$. In [FGMS26] we showed that a PVM in a Haar-random basis almost surely has no exact localization with a finite-dimensional resource and finite outcome sets; the following is the quantitative version.

Corollary 7.4 (Approximate localization). *With the universal constants c, C of Corollary 7.3, the following holds for each $d \geq 2$. Almost every rank-one PVM on $\mathbb{C}^d \otimes \mathbb{C}^d$ has a threshold $\varepsilon_0 > 0$ such that every localization with failure probability $\varepsilon < \varepsilon_0$ uses a resource of Schmidt number*

$$r \geq \max\{d, c d^{-3/5} [\ln(1/\varepsilon)]^{1/10}\}.$$

Moreover, for $0 < \varepsilon \leq 1/2$, the PVMs that have a localization with Schmidt number at most r and failure probability at most ε have outer measure at most $\min\{1, e^{C d^{10} r^{10}} \varepsilon^{\kappa_m/2}\}$.

Proof. As in [FGMS26], the parties broadcast x and y and both output $f(x, y)$. This is a protocol with free classical messages and no quantum messages, so $K_q = r$, and its two outputs always agree, so its score q_Φ equals the success probability of the localization. Apply Lemma 7.2 and Corollary 7.3. $\square$

The entanglement cost and the structure of exactly localizable measurements are studied in [PPKDG25, PBPKG26, AM26]; Corollary 7.4 is a precision-dependent bound for approximate localization of a generic basis.

8 Application: diagonal gates at fixed dimension

At fixed local dimension, Corollary 5.7 says that $\log_2 K$ must grow at least like $\log \log(1/\varepsilon)$ for almost every target, and Corollary 7.3 says the same for the quantum footprint, and hence for the number of resource qubits when only classical messages are sent. Here we show that the same lower bounds hold within the family of diagonal gates, which has Haar measure zero and is therefore not covered by the whole-group estimates.

Model and cost. In Theorem 8.1 the error is the normalized diamond distance, except for two qubits, where the Choi infidelity suffices; the reason is the restriction step in its proof. We use three cost measures: the footprint K of Section 2.3; the quantum footprint K_q of Section 7, when classical messages are free; and, when there are only classical messages (LOSCC), the number of resource qubits. Let $n_\varepsilon^{\text{res}}(U)$ be the least total number of initial resource qubits of an LOSCC protocol that implements U with normalized diamond error at most ε , where shared randomness and fresh local workspace are free and the number of qubits is bounded in every branch of the shared randomness. A pure resource of Schmidt rank r occupies at least $2 \log_2 r$ qubits.

Theorem 8.1 (Diagonal gates). *Fix $d_A, d_B \geq 2$. For almost every diagonal unitary $U|a, b\rangle = e^{i\theta_{ab}}|a, b\rangle$ on $\mathbb{C}^{d_A} \otimes \mathbb{C}^{d_B}$, with respect to the uniform measure on the phases (θ_{ab}) , there is $\varepsilon_0(U) > 0$ such that for $0 < \varepsilon < \varepsilon_0(U)$:*

- (i) *every protocol of footprint K that implements U with normalized diamond error at most ε satisfies $\log_2 K \geq \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$;*
- (ii) *every protocol with free classical messages and quantum footprint K_q that implements U with normalized diamond error at most ε satisfies $\log_2 K_q \geq \frac{1}{10} \log_2 \ln(1/\varepsilon) - O(1)$; hence $n_\varepsilon^{\text{res}}(U) \geq \frac{1}{5} \log_2 \ln(1/\varepsilon) - O(1)$.*

The $O(1)$ terms are universal. For $d_A = d_B = 2$ the same holds with the Choi infidelity in place of the diamond error, in particular for the controlled phase C_θ for almost every θ . For all sufficiently small ε , the same bounds hold for the worst case over all diagonal unitaries.

8.1 One scalar invariant

We follow the operator purity of [FGMS26], now evaluated on arbitrary, not necessarily unitary, matrices. For a complex $d^2 \times d^2$ matrix X let

$$P(X) = \frac{1}{d^4} \text{Tr}[(X \otimes X) S_A (X \otimes X)^\dagger S_A], \quad (44)$$

where S_A swaps the two copies of H_A in $(H_A \otimes H_B)^{\otimes 2}$. The proof of the operator-purity lemma of [FGMS26] applies verbatim to arbitrary matrices and shows that $P((L_A \otimes L_B)X(R_A \otimes R_B)) = P(X)$ for all local unitaries. Differentiating along $t \mapsto e^{-tb} X e^{ta}$ with $a, b \in \mathfrak{u}_{\text{loc}}$ gives

$$dP_X[-bX + Xa] = 0 \quad \text{for all matrices } X \text{ and all } a, b \in \mathfrak{u}_{\text{loc}}. \quad (45)$$

For the two-qubit controlled phase $C_\theta = \text{diag}(1, 1, 1, e^{i\theta})$, which is the case $n = 1$ of the controlled phase of [FGMS26], the same lemma gives $P(C_\theta) = (3 + \cos \theta)/4$. Since P is a sum of four-fold products of X and $X^\dagger$ and $|\text{Tr}(Y S_A Z S_A)| \leq \|Y\|_F \|Z\|_F$,

$$|dP_X[V]| \leq \frac{4}{d^4} \|X\|_F^3 \|V\|_F, \quad (46)$$

which for $d = 2$ and $\|X\|_F \leq 2$ is at most $2\|V\|_F$.

Proposition 8.2 (Controlled-phase length bound). *Let J be a compact interval in $(0, \pi)$ or in $(\pi, 2\pi)$. The set of $\theta \in J$ such that C_θ has a protocol of footprint at most K with $\delta_{\text{Choi}} \leq \varepsilon$ has Lebesgue measure at most $C_J e^{CK^2} \sqrt{\varepsilon}$. The set of $\theta \in J$ such that C_θ has a protocol with free classical messages, quantum footprint at most K_q and Choi infidelity at most ε has outer measure at most $C_J e^{CK_q^{10}} \sqrt{\varepsilon}$.*

Proof. Here $d = 2$, so the witness families of Proposition 5.1 have $D = O(K^2)$ coordinates. Every approximable C_θ has a witness $x \in \tilde{\mathcal{Z}}_\delta$ with $\|\hat{T}(x) - C_\theta\|_n \leq \delta \leq 2\sqrt{\varepsilon}$, and $\|\hat{T}\|_F \leq \|\text{Dec}\|_{\text{op}} \|\text{Enc}\|_F = 2$. By Proposition 4.2 and (45), the allowed part of the velocity does not change P , so

$$|d(P \circ \hat{T})_x(\dot{x})| = |dP_{\hat{T}}[\Xi(\dot{x})]| \leq 2\|\Xi(\dot{x})\|_F \leq 2\sqrt{h} \beta \|\dot{x}\| \leq C\delta D \|\dot{x}\|,$$

using (46), (29) and $h \leq 4\delta^2$. By (46) on the segment from $\hat{T}(x)$ to C_θ , also $|P(\hat{T}(x)) - P(C_\theta)| \leq 2\|\hat{T}(x) - C_\theta\|_F \leq 4\delta$. Consider $g(x, t) = P(\hat{T}(x)) + 4\delta t$ on $\tilde{\mathcal{Z}}_\delta \times [-1, 1]$. Every value $P(C_\theta)$ of an approximable phase lies in its image, g has bounded lifted format in $O(D)$ variables and a C^1 extension, and its derivative has norm at most $C\delta D$. Lemma 5.3 with $m = 1$, summed over the e^{CD} families, bounds the length of the set of these values by $e^{CD} \delta \leq e^{CK^2} \sqrt{\varepsilon}$. On

J the map $\theta \mapsto P(C_\theta) = (3 + \cos \theta)/4$ is monotone with derivative of absolute value at least $\min_J |\sin \theta|/4 > 0$, which proves the first claim.

For the second claim, Lemma 7.1 turns such a protocol into a protocol of footprint at most $4 \cdot 2^4 K_q^5 = 64 K_q^5$ whose score is at least as large. Substituting $K = 64 K_q^5$ proves the bound; only the score, not the whole channel, is preserved, which suffices. $\square$

Proof of Theorem 8.1. Controlled phases. Let $\varepsilon_j = e^{-j}$, $K_j = \lfloor c\sqrt{j} \rfloor$ and $K_{q,j} = \lfloor c j^{1/10} \rfloor$ with a universal $c > 0$ small enough that both bounds of Proposition 8.2 are at most $C_j e^{-j/4}$. These bounds are summable, so by the Borel–Cantelli lemma, for outer measure in the second case, almost every $\theta \in J$ admits, for all large j , no protocol with Choi infidelity ε_j and footprint at most K_j , and no protocol with free classical messages, Choi infidelity ε_j and quantum footprint at most $K_{q,j}$. Monotonicity in ε and a countable exhaustion of $(0, \pi) \cup (\pi, 2\pi)$ by intervals J give both bounds for C_θ , with the Choi infidelity, for almost every θ , and with universal constants. In LOSCC, $K_q = r$, and a protocol with shared randomness is a mixture of pure-resource protocols with at most q qubits each, so its resource has Schmidt number at most $2^{q/2}$; this gives the bound on $n_\varepsilon^{\text{res}}$. The diamond error bounds the Choi infidelity by (9).

Two qubits. Every diagonal two-qubit unitary equals, up to a global phase, $(D_A \otimes D_B) C_\theta$ with diagonal unitaries D_A, D_B and

$$\theta = \theta_{11} - \theta_{10} - \theta_{01} + \theta_{00} \pmod{2\pi}.$$

Composing the decoders with $D_A^\dagger$ and $D_B^\dagger$ changes neither the score nor the resources, and the map from the phases to θ sends the uniform measure to the uniform measure on the circle. This proves the theorem for $d_A = d_B = 2$, with the Choi infidelity.

General dimensions. Let Π_X be the projection of $\mathbb{C}^{d_X}$ onto the span of $|0\rangle$ and $|1\rangle$. Given a protocol for U , feed it only inputs in the range of $\Pi_A \otimes \Pi_B$, a local isometric embedding of $\mathbb{C}^2 \otimes \mathbb{C}^2$, and apply to each output the channel $\rho \mapsto \Pi_X \rho \Pi_X + \text{Tr}[(I - \Pi_X)\rho] \sigma_X$ for a fixed state σ_X on the two levels. Since U maps the range of $\Pi_A \otimes \Pi_B$ to itself, this gives a protocol for the two-qubit gate $\text{diag}(e^{i\theta_{00}}, e^{i\theta_{01}}, e^{i\theta_{10}}, e^{i\theta_{11}})$. Composing with channels does not increase the normalized diamond distance, and local pre- and post-processing does not change the resources. The Choi infidelity can increase under this restriction, which is why we use the diamond error in this section. Since the four phases $\theta_{00}, \dots, \theta_{11}$ are uniformly distributed, the two-qubit case gives the theorem for almost every U .

Worst case. Both bounds of Proposition 8.2 are smaller than the length of J once $K \leq c\sqrt{\ln(1/\varepsilon)}$, respectively $K_q \leq c[\ln(1/\varepsilon)]^{1/10}$, and ε is small enough. Then some C_θ with $\theta \in J$ has no protocol with these resources and Choi infidelity at most ε . By the two steps above, no diagonal unitary whose restriction to the levels 0 and 1 equals C_θ up to local diagonal unitaries and a global phase has a protocol with these resources and normalized diamond error at most ε . $\square$

Remark 8.3 (Exceptional gates). The lower bound is not pointwise. Gates with exact finite protocols have bounded cost at every precision, and they are dense in the controlled-phase circle: C_θ has an exact protocol whenever $\theta \in \pi\mathbb{Q}$ [Yu11, YGC12]. For the named gate C_1 , which has no exact protocol, Theorem C.1 in Appendix C gives only a triple-logarithmic lower bound on the number of resource qubits.

Remark 8.4 (Two-qubit gates). For general two-qubit gates the whole-group bounds apply directly: for Haar-almost every two-qubit unitary and all ε below a threshold, $\log_2 K_\varepsilon(U) \geq \frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$ by Corollary 5.7, and $n_\varepsilon^{\text{res}}(U) \geq \frac{1}{5} \log_2 \ln(1/\varepsilon) - O(1)$ by Corollary 7.3(ii), since $n^{\text{res}} \geq 2 \log_2 r$ and $K_q = r$ in LOSCC.

9 Formal verification

The Lean 4 formalization [dMU21, mC20, NLQ26] covers Theorem A and the unitary part of Theorem B, including their qubit forms, together with the qualitative score gaps and Schmidt-rank floors. The repository coverage guide lists the corresponding declarations, operational-error variants and differences between the written and formal proofs.

Assumptions. The Lean kernel accepts the proofs with only the three standard axioms of Lean’s logic (`propext`, `Classical.choice` and `Quot.sound`). The quantitative statements take three published results of real algebraic geometry as explicit hypotheses, which are stated in Lean but not proved there:

- (E1) Theorem 44 of [LRT25], the quantitative approximate definable choice used in the proof of Lemma 5.3;
- (E2) every semialgebraic set has a finite stratification, compatible with finitely many semialgebraic subsets, into semialgebraic C^1 submanifolds that are semialgebraically C^1 -diffeomorphic to open cubes and satisfy the frontier condition [Cos02, Corollary 3.8], [BCR98, Section 9.1];
- (E3) a set defined by s polynomial equations and inequalities of degree at most $\Delta \geq 2$ in k real variables has at most $\Delta(2\Delta - 1)^{k+s-1}$ connected components [Cos02, Proposition 4.13].

These are three of the results that the proof of Lemma 5.3 in Appendix A uses. The Tarski–Seidenberg projection theorem, which that proof also uses, is proved in the library from a formalization of real quantifier elimination. The Lean formulations of (E1)–(E3) have been compared by hand with Theorem 44 of the arXiv version of [LRT25] and with Corollary 3.8 and Proposition 4.13 of [Cos02]; the kernel checks only the deduction of the statements below from them.

Scope. The charged-footprint results allow arbitrary finite private and garbage registers, pure resources and common-map finite mixtures of pure resources. They use the scores of Section 2.2, with normalized diamond and joint total-variation versions for many statements; the latter Haar bounds use outer measure where measurability is not formalized. The formal PVM targets are basis matrices M rather than $M^\dagger$, an equivalent formulation by inversion invariance of Haar measure and outcome-phase invariance. Constants are universal, and almost-everywhere error thresholds depend on the target. The library also proves the main paper’s almost-everywhere exact impossibility without the three geometric hypotheses.

With free classical communication, the library proves the compression from standard Borel outcome spaces into the charged model. The floors and rates in Theorem C are formalized for finite classical alphabets, with $\frac{3}{5}n$ rather than $\frac{2}{5}n$ for universal PVM approximation. For Theorem 8.1, it proves the almost-every controlled-phase case under Choi infidelity, with finite classical alphabets for the free-classical bound and without shared randomness for the resource-qubit bound.

Limits. The precision term for PVMs in Theorem B(ii), its regional almost-everywhere consequence, and the localization corollary are not formalized at this revision. Neither are the general diagonal-gate rates, the charged-footprint worst-case diagonal bound, the shared-randomness resource-qubit claim, or the rates with standard Borel alphabets: the necessary restriction and compression lemmas exist, but are not yet combined with those conclusions. The library proves attainment of optimal scores, not of minimal diamond or total-variation errors. The image-volume and tube lemmas are formalized in the specialized forms used in the proofs. The effective arithmetic bound in Appendix C is not formalized.

10 Discussion and open questions

The exact results of [FGMS26] and the bounds of these notes come from one calculation. A frozen environment makes the compressed generators local, or diagonal for measurements, and the derivative of the overlap between the encoder and decoder isometries differs from these allowed motions only by a leakage term. At zero leakage, integration fixes a local-unitary orbit; at nonzero leakage, a bounded number of coordinates and a dimension-uniform tube estimate limit how much of the target space can be approximated. We close with some questions that this leaves open.

Fixed error and growing dimension. Our bounds concern the dependence on the precision. At constant error they only give the floors $K = \Omega(d^2)$ for universal approximation of unitaries and $K = \Omega(d^3)$ for measurements, whereas the known protocols for general unitaries use resources exponential in d . Proving super-polynomial lower bounds at fixed error remains the central open problem for NLQC and position verification [May26]; the geometric approach here would need a much finer control of the image of the protocol map than the counting of its parameters.

The dimension dependence of the precision term. Theorem A compares $O(d^2 K^2)$ parameters with $\Theta(d^4)$ transverse directions, which gives the factor d ; near SWAP and near a maximally entangled basis, Theorem B improves this to d^2 for universal approximation. We do not know whether $d^2 \sqrt{\ln(1/\varepsilon)}$ is the correct universal order, whether the almost-everywhere bounds can also be improved to d^2 , or whether the power $\sqrt{\ln(1/\varepsilon)}$ is optimal for general targets. A Hilbert–Schmidt version of the singular-value cutoff in Section 6 appears to restore the full exponent $\kappa_u/2$ in the estimate near SWAP; this would not change the universal bounds.

Free classical communication. The compression of classical transcripts in Lemma 7.1 loses a fifth power of the footprint, and with it the factors $d^{-2/5}$ and $d^{-3/5}$. The loss comes from charging the compressed transcript as a quantum message and then treating the result as an arbitrary protocol. A witness that keeps the transcript as orthogonal classical blocks, with one frozen environment and one completed decoder per transcript, appears to need only $O(d^6 K_q^6)$ coordinates, which would improve the exponent $\frac{1}{10}$ to $\frac{1}{6}$; we have not written this out. Removing the loss entirely would bring the bounds with free classical communication close to those with charged messages.

Entropy. All our bounds concern Schmidt numbers and message dimensions, and they give no lower bound on the entanglement entropy of the resource. It remains open whether the targets excluded in [FGMS26] can be implemented exactly with a resource of infinite Schmidt rank but finite entanglement entropy, together with finite-dimensional messages.

Explicit targets. For C_1 , Appendix C gives $\log_2 K = \Omega(\log \log \log(1/\varepsilon))$, while almost every controlled phase requires $\log_2 K = \Omega(\log \log(1/\varepsilon))$ (Theorem 8.1). Closing this gap requires either a better effective arithmetic argument or a different lower-bound method for named targets.

Other models. The whole-group results of Sections 5 to 7 are stated for balanced local dimensions; the absorption of dimension factors in the proof of Theorem 5.5 is not uniform in the aspect ratio d_A/d_B , and unbalanced dimensions need a separate argument. The exact argument is expected to extend to any fixed number of parties [FGMS26]; a quantitative multipartite theorem would in addition need a suitable resource measure and a joint compression of the

environment. For general channels, and for rank-one POVMs, whose analysis operators have additional directions of motion, the frozen dilation is weaker and the argument has to be modified.

Position verification. Our volume bound, Theorem A(i), averages over targets, while security of position verification requires lower bounds for the verifier’s actual task and success criterion, typically at nonzero error on a fixed input distribution. The scores we use are of this form for the uniform input distribution on a basis or on half of a maximally entangled state, but the targets of position-verification protocols are specific and are not covered by the almost-everywhere statements.

AI statement

These notes are currently an almost-entirely AI-written exposition, reporting explorations of possible extensions to the results of [FGMS26]. Various models, including OpenAI GPT-6 Astra, Anthropic Claude Fable 5.1, and Opus 5.5 were involved in proving the results and creating these notes, in August and September 2026, following the authors’ direction and selection.

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A Proof of the tube estimate

In this appendix all matrix norms are ordinary Frobenius norms, so $\mathbb{C}^{d^2 \times d^2} \cong \mathbb{R}^{2N}$ with $N = d^4$. We write ω_m for the volume of the unit ball in $\mathbb{R}^m$ and $U_\tau(E)$ for the closed Frobenius τ -neighbourhood of a set E . The argument has three parts: a Jacobian bound for semialgebraic images (Lemma 5.3), a comparison of ambient volume with Haar measure (Lemma A.1), and the application to a thickened image.

A.1 Proof of the image-volume lemma

Proof of Lemma 5.3. If $k < m$ the image has m -volume zero, because ϕ is locally Lipschitz, so assume $k \geq m$ and put $n_{\text{tot}} = k + m + j$.

Approximate choice reduces the source dimension. Apply Theorem 44 of Lerario, Rizzi and Tiberio [LRT25] to a compact lift S of the graph of $\phi|_Y$, with the projection onto the m output coordinates. For every $\tau > 0$ it gives compact semialgebraic sets A_τ in the τ -neighbourhood of S , of dimension at most m , whose output projections converge to $\phi(Y)$ in Hausdorff distance, and the numbers and degrees of the polynomials describing A_τ are bounded independently of n_{tot} and τ . Let C_τ be the projection of A_τ to the source coordinates. Then

$$\dim C_\tau \leq m, \quad C_\tau \subseteq U_\tau(Y), \quad d_H(\phi(C_\tau), \phi(Y)) \rightarrow 0 \quad (\tau \rightarrow 0).$$

The last statement follows from uniform continuity of the fixed extension of ϕ : the source and output coordinates of a point near the lifted graph satisfy the graph equation up to an error that tends to zero.

Coordinate projections bound the size of the choice. For every affine plane E of codimension m in the source,

$$b_0(E \cap C_\tau) \leq C_\Delta^{n_{\text{tot}}}, \quad (47)$$

where b_0 counts connected components. Indeed, impose the equations of E on A_τ before projecting away the other coordinates; their sum of squares is a single quadratic equation, so the component bound for semialgebraic sets of fixed format in n_{tot} variables applies [BCR98, BPR06], and projection does not increase the number of components.

Stratify C_τ into smooth semialgebraic pieces. At an m -dimensional tangent plane, let j_I be the Jacobian of the projection onto a set I of m coordinate axes. The Cauchy–Binet formula gives $\sum_I j_I^2 = 1$, hence $\sum_I j_I \geq 1$. For almost every point of a coordinate m -plane, the fibre of the projection in C_τ is finite, with at most $C_\Delta^{n_{\text{tot}}}$ points by (47), and lower-dimensional and critical images have m -volume zero. The area formula [Fed69, 3.2.20] therefore gives, for $\tau < 1$,

$$\mathcal{H}^m(C_\tau) \leq \sum_I \int_{C_\tau} j_I d\mathcal{H}^m \leq \binom{k}{m} C_\Delta^{n_{\text{tot}}} \omega_m (R_0 + 1)^m \leq C'_\Delta^{n_{\text{tot}}} \omega_m.$$

All dimension-dependent factors have thus been absorbed into an exponential in n_{tot} .

Jacobian bound and limit. Apply the area formula to ϕ on the same strata. Its restricted Jacobian is at most its ambient m -Jacobian, so

$$\text{Vol}_m(\phi(C_\tau)) \leq C'_\Delta^{n_{\text{tot}}} \omega_m \left(\sup_Y J_m \phi + o(1) \right),$$

where the $o(1)$ term comes from the continuity of the ambient derivative on $U_\tau(Y)$. For fixed dimensions, the sets C_τ , their images under the fixed extension of ϕ , and their descriptions have uniformly bounded format and lie in a fixed ball. With $\delta_\tau = d_H(\phi(C_\tau), \phi(Y)) \rightarrow 0$, the continuity of volume for tame sets [YC04, Theorem 5.10] gives

$$|\text{Vol}_m(\phi(C_\tau)) - \text{Vol}_m(\phi(Y))| \leq C_*(\delta_\tau + \delta_\tau^m) \rightarrow 0.$$

The constant C_* may depend on the dimensions and on the extension; it multiplies a vanishing quantity and does not enter the final constant. Letting $\tau \rightarrow 0$ proves the lemma. $\square$

The use of Theorem 44 of [LRT25] keeps the dependence on the dimensions explicit; the packaged variation estimate in their Theorem 45 does not by itself give the dimension-independent base needed here. Only the graph on Y needs a lift of bounded format; the extension away from Y is used for continuity and differentiation.

A.2 The volume of the unitary group at its natural scale

The unitary group $U(d^2)$ has Frobenius diameter of order d . The following lemma compares the volume of a neighbourhood of a subset with its Haar measure at this scale.

Lemma A.1 (Normal volume). *Let $d \geq 2$ and $N = d^4$. There are universal $c, C > 0$ such that $\text{Vol}_F U(d^2) \geq (cd)^N \omega_N$, and for every measurable $\mathcal{S} \subseteq U(d^2)$ and $0 < t \leq d/64$,*

$$\text{Vol}_{2N}(U_t(\mathcal{S})) \geq c^N \omega_N t^N \text{Vol}_F U(d^2) \mu(\mathcal{S}). \quad (48)$$

Proof. First, at least a quarter of the Hermitian directions Q of Frobenius norm one satisfy $\|Q\|_{\text{op}} \leq 32/d$. To see this, let G be a standard Gaussian vector in the N -dimensional real space of Hermitian $d^2 \times d^2$ matrices. For every complex unit vector v , $v^\dagger G v$ is a standard real Gaussian. A $\frac{1}{4}$ -net of the complex unit sphere of $\mathbb{C}^{d^2}$ has at most 9^{2d^2} points, and $\|G\|_{\text{op}} \leq 2 \max_v |v^\dagger G v|$ over the net. Hence $\Pr(\|G\|_{\text{op}} > 16d) \leq 2 \cdot 9^{2d^2} e^{-32d^2} < \frac{1}{4}$. For $X = \|G\|_F^2$ we have $\mathbb{E}X = N$ and $\mathbb{E}X^2 = N(N+2)$, and the Paley–Zygmund inequality gives $\Pr(X \geq N/4) \geq \frac{9}{16} \frac{N}{N+2} \geq \frac{1}{2}$ since $N \geq 16$. By the union bound, both events occur with probability at least $1 - \frac{1}{4} - \frac{1}{2} = \frac{1}{4}$, and then $\|G\|_{\text{op}}/\|G\|_F \leq 32/d$. Since the direction of G is uniform, this proves the claim, and consequently

$$\text{Vol}\{Q = Q^\dagger : \|Q\|_F < t, \|Q\|_{\text{op}} \leq \frac{1}{2}\} \geq \frac{1}{4} \omega_N t^N \quad (t \leq d/64).$$

For such Q , the map $(U, Q) \mapsto U(I+Q)$ is injective by uniqueness of the polar decomposition, and $\|U(I+Q) - U\|_F = \|Q\|_F$. After left translation its differential is $(A, Q') \mapsto A + Q' + AQ$, with A anti-Hermitian and Q' Hermitian. The map $(A, Q') \mapsto A + Q'$ is an isometry, and the perturbation has norm at most $\|A\|_F/2$, so the Jacobian is at least 4^{-N} . Integrating over $\mathcal{S}$ and the set of Q above proves (48).

For the volume of the group, use the Cayley chart $A \mapsto (I+A)(I-A)^{-1}$ on anti-Hermitian A with $\|A\|_{\text{op}} \leq 1$. It is injective, and its derivative $B \mapsto 2(I-A)^{-1}B(I-A)^{-1}$ has all singular values at least $\frac{1}{2}$. By the same angular estimate, at least a quarter of the Frobenius ball of radius $d/32$ in the anti-Hermitian matrices lies in the domain, so the image has volume at least $2^{-N} \cdot \frac{1}{4} \omega_N (d/32)^N$. $\square$

A.3 Thickening the image

Proof of Proposition 5.4. Use ordinary Frobenius norms and put

$$\Theta_s(x, Y') = \hat{T}(x) + 2ds Y', \quad x \in \tilde{\mathcal{Z}}, Y' \in \mathbb{C}^{d^2 \times d^2}, \|Y'\|_F \leq 1.$$

The normalized bounds L and λ become the ordinary bounds dL and $d\lambda$. Since $d\Theta_s d\Theta_s^* = d\hat{T} d\hat{T}^* + 4d^2 s^2 I$, the singular values of $d\Theta_s$ are at most $\sigma_j(d\hat{T}) + 2ds$, so

$$J_{2N}\Theta_s \leq d^{2N} (L + 2s)^\ell (\lambda + 2s)^{2N-\ell}. \quad (49)$$

The source $\tilde{\mathcal{Z}} \times \{\|Y'\|_F \leq 1\}$ has universal radius and a graph lift of bounded format with $O(D+N)$ variables. The image of Θ_s is the neighbourhood $U_{2ds}(\hat{T}(\tilde{\mathcal{Z}}))$, so Lemma 5.3 gives

$$\text{Vol}_{2N}(U_{2ds}(\hat{T}(\tilde{\mathcal{Z}}))) \leq e^{C\Delta(D+N)} \omega_{2N} d^{2N} (L + 2s)^\ell (\lambda + 2s)^{2N-\ell}.$$

Every point of $\mathcal{S}$ is within Frobenius distance $d\tau \leq ds$ of $\hat{T}(\tilde{\mathcal{Z}})$, so $U_{ds}(\mathcal{S}) \subseteq U_{2ds}(\hat{T}(\tilde{\mathcal{Z}}))$. Lemma A.1 with $t = ds \leq d/64$, that is $s \leq c_0 := 1/64$, gives

$$\text{Vol}_{2N}(U_{ds}(\mathcal{S})) \geq c^N \omega_N^2 d^{2N} s^N \mu(\mathcal{S}).$$

Now $\omega_{2N} \leq \omega_N^2$, $\lambda \leq s$, and the factors d^{2N} cancel. The remaining power of s is $s^{2N-\ell}/s^N = s^{N-\ell}$, which proves the proposition. A union of e^{CD} sets changes the constant in the exponential only. $\square$

B Schmidt-rank floors

These elementary bounds verify the size conditions on K used in Corollaries 5.6, 6.4 and 6.8, and they record constant-error bounds that do not depend on the tube estimate. For exact unitaries, the first is the familiar requirement that the available Schmidt rank must cover the operator-Schmidt rank of the gate; Stahlke and Griffiths prove the resource-only version for separable operations [SG11]. In NLQC, quantum messages also carry Schmidt rank across the cut, which is why the bound is in terms of $K = r m_A m_B$.

Lemma B.1 (Schmidt-rank floors). *Let a protocol of footprint at most K have score deficit at most $\varepsilon < 1$.*

(i) *For a unitary target U , $\Lambda_K(|U\rangle\rangle) \geq 1 - \varepsilon$, where $|U\rangle\rangle$ is split by the laboratory cut.*

(ii) *For a PVM with basis $(\varphi_i)_{i \leq d^2}$, there are integers $k_i \geq 0$ with*

$$\sum_i k_i \leq K, \quad \sum_i \Lambda_{k_i}(\varphi_i) \geq d^2(1 - \varepsilon)^2.$$

Consequently, approximating SWAP requires $K \geq d^2(1 - \varepsilon)$, every basis requires $K \geq d^2(1 - \varepsilon)^2$, and a complete basis of maximally entangled vectors requires $K \geq d^3(1 - \varepsilon)^2$. At $\varepsilon = 0$, (i) gives that K is at least the operator-Schmidt rank of U , and (ii) gives $K \geq \sum_i \text{SR}(\varphi_i)$. Moreover:

(iii) *If U has full operator-Schmidt rank d^2 , which holds for Haar-almost every U , then $K \geq d^2$ whenever $\varepsilon < \varpi_{\min}(U)$, the smallest squared Schmidt coefficient of $|U\rangle\rangle$. If every basis vector φ_i has Schmidt rank d , which holds for almost every PVM, then $K \geq d^3$ whenever $\varepsilon < \lambda_*/(2d^2)$, where λ_* is the smallest squared Schmidt coefficient of all the φ_i .*

Proof. Select the pure resource component and use the notation of the proof of Lemma 3.2. (i) There, $1 - \varepsilon \leq p \leq \Lambda_K(|U\rangle\rangle \otimes \gamma') \leq \Lambda_K(|U\rangle\rangle)$ by (12). (ii) There, $\Lambda_K(v_0) \geq p^2 \geq (1 - \varepsilon)^2$. Let k_i be the number of selected triples (i, j, k) in block i of (17), and let $\mathcal{J}_i$ be the set of indices j that occur in them, so $|\mathcal{J}_i| \leq k_i$. Since $\sum_k \zeta_{ik}^2 = \|\tilde{\gamma}_i\|^2 \leq 1$, the selected mass in block i is at most $d^{-2} \sum_{j \in \mathcal{J}_i} \lambda_{ij} \leq d^{-2} \Lambda_{k_i}(\varphi_i)$. Summing over i proves (ii).

The normalized operator-Schmidt spectrum of SWAP consists of d^2 equal weights, so $\Lambda_K(|\text{SWAP}\rangle\rangle) = \min\{K/d^2, 1\}$. For every unit vector, $\Lambda_0 = 0$ and $\Lambda_k \leq 1$, so $\sum_i \Lambda_{k_i}(\varphi_i) \leq |\{i : k_i \geq 1\}| \leq \sum_i k_i \leq K$. A maximally entangled vector has $\Lambda_k \leq k/d$, so $\sum_i \Lambda_{k_i}(\varphi_i) \leq K/d$.

For (iii), if $K < d^2$ then $\Lambda_K(|U\rangle\rangle) \leq 1 - \varpi_{\min}(U)$, so (i) forces $\varepsilon \geq \varpi_{\min}(U)$. If $\sum_i k_i < d^3$, then $k_i < d$ for some i , so $\sum_i \Lambda_{k_i}(\varphi_i) \leq d^2 - \lambda_*$, and (ii) forces $(1 - \varepsilon)^2 \leq 1 - \lambda_*/d^2$, which fails for $\varepsilon < \lambda_*/(2d^2)$. The determinant of the realignment of U , and the product over i of the determinants of the $d \times d$ coefficient matrices of the columns of a basis matrix, are real-analytic functions on $U(d^2)$ that do not vanish identically (for instance at SWAP and at a basis of maximally entangled vectors), so their zero sets are null. $\square$

For $0 < \varepsilon \leq 1/2$, universal approximation of unitaries thus forces $K \geq d^2/2$, and universal approximation of PVMs forces $K \geq d^2/4$ already because of a single basis, and in fact $K \geq d^3/4$. The precision term $cd\sqrt{\ln(1/\varepsilon)}$ of Theorem A exceeds these floors only when $\ln(1/\varepsilon)$ is of order d^2 for unitaries and d^4 for measurements, while the term $cd^2\sqrt{\ln(1/\varepsilon)}$ of Theorem B adds to the floor d^2 for unitaries at every error, and exceeds the floor d^3 for measurements once $\ln(1/\varepsilon)$ is of order d^2 .

C An effective bound for an explicit controlled-phase gate

The main paper [FGMS26] proves that $C_1 = \text{diag}(1, 1, 1, e^i)$ has no exact protocol with finite-dimensional resources: its operator purity $P(C_1) = (3 + \cos 1)/4$ is transcendental, whereas the

purity of every exactly implementable unitary is algebraic. By Proposition 2.2, the least error $\varepsilon_K(C_1)$ at footprint K is positive, but that compactness argument gives no rate. The almost-everywhere results also give no bound for this named target. Here we make the arithmetic obstruction effective, using quantifier elimination and a transcendence measure. Throughout this appendix, $d = 2$.

Theorem C.1 (Effective bound for C_1). *There are effectively computable constants $C_E, C_{E,q} > 0$ with the following property. For $K \geq 1$ and $0 < \varepsilon < 1$, every protocol of footprint at most K that implements C_1 with $\delta_{\text{Choi}} \leq \varepsilon$ satisfies*

$$\varepsilon \geq \exp[-\exp(C_E K^2)]. \quad (50)$$

If classical messages are free as in Section 7 and $K_q \geq 1$, then $\varepsilon \geq \exp[-\exp(C_{E,q} K_q^{10})]$. Consequently, for $0 < \varepsilon < 1/e$,

$$\log_2 K_\varepsilon(C_1) \geq \frac{1}{2} \log_2 \ln \ln(1/\varepsilon) - O(1), \quad \log_2 K_{q,\varepsilon}(C_1) \geq \frac{1}{10} \log_2 \ln \ln(1/\varepsilon) - O(1),$$

where $K_{q,\varepsilon}$ is the least quantum footprint; in LO SCC this bounds the Schmidt number of the resource.

Proof of Theorem C.1. For $-1 \leq c \leq 1$, $s \geq 0$ and $c^2 + s^2 = 1$, let $C(c, s) = \text{diag}(1, 1, 1, c + is)$, so that $C_1 = C(c_*, s_*)$ with $c_* = \cos 1$ and $s_* = \sin 1$.

Step 1: the least deficit as a compact polynomial optimization. Fix a factorization $r m_A m_B \leq K$. The score is affine in the resource, so it suffices to consider pure resources on $\mathbb{C}^r \otimes \mathbb{C}^r$. By Lemma 4.1, the kept register of party X has dimension at most $2r m_X$, and the input of its final channel has dimension at most $2K$. The encoders are isometries. At the end we discard everything except the logical output, and represent each final channel, from a space of dimension at most $2K$ to the fixed output $\mathbb{C}^2$, by a Stinespring isometry with environment of dimension at most $4K$. The resource, the two encoders and the two final isometries then use

$$N_{\text{var}} \leq CK^2 \quad (51)$$

real coordinates. For each choice of the finitely many discrete dimensions, the normalization and isometry equations define a compact real algebraic set $\mathcal{Z}$ over $\mathbb{Q}$, the analogue of the strategy sets $\mathcal{Z}_\nu$ of [FGMS26] without the exactness constraint. In real coordinates, the score $q_{C(c,s)}$ is a polynomial with rational coefficients of fixed degree in (c, s, x) . We package all constraints for one dimension choice into one sum of squares; with the fixed list of score and box inequalities, the number and degrees of the polynomials are fixed, and after clearing denominators their integer coefficients have bit size polynomial in K . Let $\mathcal{F}_K$ be the resulting finite collection, and put $f = 1 - q_{C(c,s)}$ and

$$g_K(c) = \min_{\mathcal{Z} \in \mathcal{F}_K} \min\{f(c, s, x) : x \in \mathcal{Z}, s \geq 0, c^2 + s^2 = 1\}. \quad (52)$$

All minima are attained, and $0 \leq g_K \leq 1$. All reductions above are explicit, so the constants in these format bounds can be computed.

A protocol with score one implements its target exactly (Proposition 2.2), and C_1 has no exact protocol with finite-dimensional resources [FGMS26]. Hence

$$g_* := g_K(c_*) > 0,$$

and a protocol of footprint at most K with $\delta_{\text{Choi}} \leq \varepsilon$ for C_1 gives $g_* \leq \varepsilon$.

Step 2: two effective inputs. One-block quantifier elimination over the reals turns an existential formula with N_{var} quantified variables and two free variables, built from a fixed

number of integer polynomials of degree at most d_0 and coefficient bit size at most τ , into a quantifier-free formula whose polynomials Q satisfy

$$\deg Q \leq d_0^{O(N_{\text{var}})}, \quad \text{bitsize}(Q) \leq \tau d_0^{O(N_{\text{var}})} \quad (53)$$

[BPR06, Theorem 14.16]. In general these exponents also grow with the number of free variables and of quantifier blocks; here there is a single block and the number of free variables is fixed at two, so only the dependence on N_{var} remains. Cijssouw's transcendence measure for e^β [Cij74, Theorem 1] implies that for every nonzero algebraic β there are effectively computable $C_\beta, c_\beta > 0$ such that every nonconstant integer polynomial S of degree ℓ and coefficient height H satisfies

$$|S(e^\beta)| \geq \exp[-C_\beta (\ell + \ln H)^{c_\beta}]. \quad (54)$$

The precise exponents are not needed; any bound of this polynomial type gives the same final form.

Step 3: an algebraic boundary near the target. Choose $\mathcal{Z}_* \in \mathcal{F}_K$ that attains g_* , and consider the epigraph

$$E = \{(c, y) \in [-1, 1] \times [0, 1] : \exists s \geq 0, x \in \mathcal{Z}_* \text{ with } c^2 + s^2 = 1, f(c, s, x) \leq y\}.$$

The sine coordinate is quantified out, and only c is kept. By (51) and (53), E is a Boolean combination of sign conditions on nonzero integer polynomials $Q(c, y)$ of degree L and bit size τ_Q with

$$L, \tau_Q \leq \exp(A_0 K^2) \quad (55)$$

for an effective constant A_0 . The point (c_*, g_*) lies on the boundary of E : it belongs to E , while $(c_*, g_* - t) \notin E$ for $0 < t < g_*$. Hence some polynomial Q of the formula vanishes at (c_*, g_*) ; otherwise all signs, and hence membership in E , would be locally constant. Divide Q by the largest power of y that divides it; since $g_* > 0$, the quotient still vanishes at (c_*, g_*) , and we rename it Q . Then $Q_0(c) := Q(c, 0)$ is a nonzero integer polynomial.

Step 4: transcendence separates the target from the boundary. Let Q_0 have degree $\ell \leq L$ and height $H \leq H_Q \leq 2^{\tau_Q}$. If $\ell \geq 1$, then $S(z) = 2^\ell z^\ell Q_0(\frac{z+z^{-1}}{2})$ is a nonconstant integer polynomial of degree at most 2ℓ and height at most $(\ell + 1)2^\ell H$, and $|Q_0(\cos 1)| = 2^{-\ell} |S(e^i)|$. Applying (54) with $\beta = i$, and noting that a nonzero constant Q_0 has absolute value at least one,

$$|Q_0(c_*)| \geq \exp[-C'(\ell + \ln H + 1)^{c'}].$$

On the other hand, the mean value theorem in y gives

$$|Q_0(c_*)| = |Q(c_*, 0) - Q(c_*, g_*)| \leq g_*(L + 1)^3 H_Q,$$

where the last factor bounds $|\partial_y Q|$ on $[-1, 1] \times [0, 1]$. Combining these with (55) gives an effective $C_E > 0$ with $g_* \geq \exp[-\exp(C_E K^2)]$, and hence (50).

Step 5: free classical messages. By Lemma 7.1 with $d = 2$, a protocol with quantum footprint K_q yields a protocol of footprint at most $\bar{K} = 64K_q^5$ whose score is at least as large. Hence $\varepsilon \geq g_{\bar{K}}(c_*) \geq \exp[-\exp(C_E \bar{K}^2)] \geq \exp[-\exp(C_{E,q} K_q^{10})]$, using that g_K is nonincreasing in K . Taking logarithms twice gives the stated bounds on $\log_2 K_\varepsilon(C_1)$ and $\log_2 K_{q,\varepsilon}(C_1)$. $\square$

Real quantifier elimination has been used before to show decidability of compact, fixed-dimensional questions in quantum information [WCPG11]; here the explicit degree and height bounds are what make the gap effective.

The bound is weaker than the generic result of Theorem 8.1: for almost every controlled phase, $\log_2 K_\varepsilon$ grows at least as $\frac{1}{2} \log_2 \ln(1/\varepsilon) - O(1)$, whereas the explicit bound has the additional logarithm. In LOSCC this gives a triple-logarithmic lower bound on the total

resource-qubit cost, since a resource of Schmidt number r occupies at least $2 \log_2 r$ qubits. It does not give a lower bound on entanglement entropy.

The same proof applies to C_θ for every nonzero real algebraic θ , with constants depending on θ : the exact obstruction uses Hermite–Lindemann, and Cijssouw’s measure applies with $\beta = i\theta$. If $\sin \theta < 0$, complex conjugating the target and all protocol maps preserves the score and the footprint and reduces to the upper semicircle used in the proof.